

Controlled Substance Designation and Medical Substitution: Evidence from Gabapentin Monitoring and Upscheduling

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Abstract

A key concern in U.S. substance use policy is whether supply-side controls can effectively reduce inappropriate prescribing and associated harms, without leading to substitution towards more dangerous substances. Gabapentin, a prescription anticonvulsant approved for nerve pain and seizures, is increasingly used off-label for conditions like postoperative pain, alcohol withdrawal, and anxiety. However, mounting evidence of gabapentin misuse and related harms, particularly when combined with opioids, has led several U.S. states to classify it as a Schedule V controlled substance under the Controlled Substances Act (CSA) and to mandate its prescription monitoring through state prescription drug monitoring programs (PDMPs). This study evaluates the impact of tighter prescribing controls on gabapentin following its controlled substance designation, focusing on misuse and related harms in states that adopted this policy. Using a novel, comprehensive dataset that integrates National Vital Statistics multiple cause of death certificate records and proprietary MarketScan claims data—we employ a staggered difference-in-differences framework to assess the policy’s effects. We find that classifying gabapentin as a *drug of concern*, which entails inclusion in PDMPs but does not restrict prescribing, had little to no impact on prescribing rates. In contrast, Schedule V classification, which introduces stricter controls and documentation requirements, resulted in a sustained and statistically significant reduction in the share of patients receiving gabapentin prescriptions. We find limited evidence of clinically concerning substitution behavior in treatment. There is no detectable increase in prescribing of opioids, and co-prescribing of gabapentin and opioids declines following scheduling.

Keywords: .

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1 Introduction

The U.S. opioid epidemic has intensified scrutiny of pain management and drug regulation, raising questions about the unintended effects of supply-side policies. Restricting one class of medications can spur shifts to others; for instance, constraining OxyContin access contributed to a surge in heroin use (Alpert et al., 2018). Amid these regulatory shifts, gabapentin – an anticonvulsant – has seen dramatic growth in use and although gabapentin is not federally scheduled, its rising misuse and frequent co-prescription with opioids have drawn considerable attention and concern (Powell et al., 2023; Lavin et al., 2021; Torrance et al., 2020; Pauly et al., 2020; Hah et al., 2018). Gabapentin can potentiate opioid effects and, in some cases, combinations have been lethal (Powell et al., 2023), prompting calls for tighter regulations and scheduling that is comparable to opioids with high potential for misuse. Given its increasing off-label use for a range of conditions beyond its original indications, regulating gabapentin may alter treatment pathways and affect downstream health outcomes. This paper examines the impact of state-level efforts to monitor and control gabapentin prescribing, focusing on misuse, potential substitution to both other medications and non-pharmacological treatments, and mortality.

Gabapentin is now commonly prescribed off-label for various conditions, including migraines, some mental health disorders, insomnia, and alcohol withdrawal symptoms (Peckham et al., 2018; Modesto-Lowe et al., 2019). It is also increasingly used as an opioid-sparing agent in chronic pain management (Peckham et al., 2018). However, the evidence supporting these uses is generally weak, given that gabapentin is found to provide modest benefits (if any) for non-epileptic pain or mood disorders (Peckham et al., 2018). This extensive off-label use has made gabapentin one of the most prescribed medications in the U.S. – ranking seventh overall – largely for indications beyond its FDA labels (Powell et al., 2024). These prescribing patterns have occurred amid growing reports of abuse liability: individuals with substance-use histories report that gabapentin can elicit opioid-like euphoria, dissociative sensations, and enhanced sedation, especially at higher doses (Smith et al., 2016). Recent work also finds that co-administration of gabapentinoids (such as gabapentin)¹ with opioids nearly doubles the hazard of opioid overdose compared to opioids alone (Bykov et al., 2020). Given these concerning trends, the FDA and CDC have issued warnings that gabapentin can cause serious, life-threatening respiratory depression, especially when combined with opioids or other

¹Gabapentinoids include gabapentin, pregabalin, and mirogabalin.

central nervous system depressants (Mattson et al., 2022; U.S. Food and Drug Administration, 2019). Using data from the CDC’s National Vital Statistics System mortality data, Figure A1 shows a stark increase in the number of deaths where gabapentinoids are listed as a cause of death, and notably, 80% of these also involved an opioid. There is also evidence of physiological dependence: case reports document that tolerance and withdrawal symptoms can occur upon abrupt discontinuation of gabapentin (Smith et al., 2016). Taken together, emerging evidence indicates that gabapentin, once considered relatively benign, may carry meaningful risks even at approved doses—particularly when used in combination with opioids or other sedating medications (Bykov et al., 2020; U.S. Food and Drug Administration, 2019).

The concerning rise in gabapentin prescribing we have outlined has led to broadly two state-level policy responses. One, some states have classified or upscheduled the drug as a Schedule V controlled substance—despite its unscheduled status federally—thereby imposing stricter rules on prescribing, such as refill limits and mandatory prescription drug monitoring. Grauer and Cramer (2022) find that these policies are associated with a reduction in the days supplied of gabapentin among Medicare Part D patients. When examining the upscheduling of a specific opioid, Tramadol, Gupta et al. (2023) finds that this change leads to a substitution to other opioids and thus no net change in opioid prescribing. Therefore, it is unclear whether gabapentin upscheduling can lead to harmful substitution or other prescribing practices, including the use of opioids. A possible concern with a schedule V designation, is that in addition to various restrictive opioid policies, that such regulation of gabapentin may reduced appropriate access to patients with need for this type of pharmaceuticals. Two, rather than formally up-scheduling gabapentin, some states have designated it as a *drug of concern*, requiring its monitoring through state prescription drug monitoring programs (PDMPs).² These distinct policy responses allow us to examine how health care prescribing and utilization differ under a monitoring-only regime versus a stricter Schedule V classification. This distinction is crucial for policymakers and clinicians: although both interventions aim to prevent misuse, they differ significantly in their regulatory requirements and clinical implications. Evaluating how these policy interventions shape prescribing behavior, treatment alternatives, and patient outcomes can inform state efforts to promote safe prescribing while maintaining access to effective care.

²PDMPs are government-operated electronic systems designed to track the prescribing and dispensing of controlled substances. They aim to reduce misuse, prevent “doctor shopping,” and inform clinical practice.

In this paper, we exploit the staggered timing of state Schedule V classification and, less stringent, PDMP monitoring of gabapentin on prescribing and patient outcomes within the healthcare system. We examine (1) gabapentin prescribing itself and how it changes after scheduling; (2) prescribing of opioids and other analgesics, including the rate of co-prescribing gabapentin with opioids; and (3) potential substitution to alternative pain treatments or medications (e.g., tramadol or pregabalin). In particular, we look for spillovers into physical and mental healthcare; for instance, whether patients shift toward nonpharmacologic management, such as physical therapy. Finally, we assess any downstream effects on overdose mortality.

Thus far, we find that Schedule V classification of gabapentin leads to an immediate and lasting decrease in prescribing of gabapentin. We find no effect of the drug of concern designation on prescribing. We also find no effect on spillovers to opioid prescribing. Together, this finding suggests a decrease in problematic co-prescribing of opioids and gabapentin. We also find that Schedule V classification leads a decrease in mortality involving both gabapentinoids and opioids.

2 Background and Conceptual Model

2.1 Regulatory Context and Motivation

Unlike medications like prescription opioids, gabapentin was long perceived as having a low abuse profile. However, rising overdose deaths involving gabapentin, alongside evidence that it may potentiate opioid effects and respiratory depression (Bykov et al., 2020; Mattson et al., 2022), have raised serious concerns. In response, several states have adopted policies to increase oversight of gabapentin prescribing. These include: (1) monitoring-only approaches, wherein gabapentin is designated as a “drug of concern” and tracked in state prescription drug monitoring programs (PDMPs); and (2) formal scheduling, where the drug is reclassified as a Schedule V controlled substance under state controlled substances acts, triggering the full set of controlled-substance requirements, including mandatory PDMP queries, refill limits, and enhanced recordkeeping. These regulatory responses differ significantly in both stringency (Table 1 summarizes prescribing restrictions by regulatory classification), and therefore their potential implications for provider behavior.

Federal and state agencies may initially designate medications like gabapentin as “drugs of concern” as a lower-cost, lower-burden regulatory response while evidence on

misuse is still accumulating. This approach allows the drug to be monitored through Prescription Drug Monitoring Programs (PDMPs) without invoking the legal and administrative processes required for formal scheduling. Under the Controlled Substances Act (CSA), federal scheduling requires a multi-step evaluation involving the Food and Drug Administration (FDA) and Drug Enforcement Administration (DEA), including an eight-factor analysis of abuse potential, public health risk, and medical utility (21 U.S.C. §811). This process can be time-consuming, politically sensitive, and dependent on robust epidemiologic and toxicologic data. Moreover, scheduling a widely used medication like gabapentin, which has recognized therapeutic value for chronic pain and neurological conditions, raises concerns about limiting access for legitimate patients, increasing prescriber burden, and potentially driving substitution toward more dangerous alternatives. As such, designating gabapentin as a drug of concern provides a regulatory middle ground that signals risk and allows for data collection without triggering the full constraints of a controlled substance classification (Karavolis et al., 2021; Drug Enforcement Administration, 2020). At the state level, this designation also offers more flexibility and responsiveness, as state PDMP boards or health departments often have more discretion to flag emerging substances of concern without legislative or rule making delays.

The staggered designation of DOC or C-V controlled substance status across states and over time generate plausibly exogenous variation in prescribing environments that we use to examine how regulatory frictions and information affect provider behavior.

2.2 Related Literature

Our analysis builds on several strands of research on behavioral responses to health policy.

First, we relate to work showing that policy signals can alter provider behavior through changes in perceived risk or scrutiny, even without new information—for example, following FDA black-box warnings. DOC designation or scheduling may increase prescriber perception of the risks associated with gabapentin use, thereby negatively impacting the prescriber’s perception of the net gain from the medication for the patient, and increasing a sense of more dire consequences if now found negligent through prescriber monitoring. Previous research on FDA drug withdrawals, FDA black-box warnings and Consumer Production Safety Commission non-prescription product (children’s toys) recalls shows that competitors in the same therapeutic class could experience

negative spillovers in some settings (Parkinson et al., 2014; Collins et al., 2013; Freedman et al., 2012; Cawley and Rizzo, 2008). We refer to this as a *salience channel*, in which regulatory labeling increases the perceived cost of prescribing.

Second, a large body of literature emphasizes the informational role of PDMPs in shaping prescribing (Bao et al., 2016; Buchmueller and Carey, 2018; Meara et al., 2016). PDMPs reveal patient-specific controlled-substance histories, allowing prescribers to update beliefs about misuse risk. PDMP inclusion alone may increase transparency and create soft deterrents to inappropriate prescribing, as providers become aware that prescribing patterns are being tracked. However, it imposes minimal legal restrictions on access. Furthermore, in our setting, DOC designation combined with mandatory PDMP access generates a clear test for whether the provision of new information—as distinct from heightened salience alone—drives reductions in prescribing.

Third, we connect to work on *hassle* costs in health policy (Finkelstein et al., 2019; Alpert et al., 2024; Sacks et al., 2021). Schedule V classification introduces binding legal constraints such as refill limits, written prescription requirements, and enhanced penalties for noncompliance. Such administrative burdens can reduce utilization even when demand remains unchanged. In canonical models, ordeal costs screen out low-benefit users by raising participation costs. In our context, however, these administrative frictions apply uniformly to prescribers rather than selectively to patients, offering no screening value. Hence, ordeal and hassle costs are empirically indistinguishable, and we treat them jointly as a *hassle-cost* channel reflecting fixed administrative burden.

Fourth, our study contributes to the literature on unintended consequences of supply side constraints on prescribing of pain medication, particularly implications for harmful substitution behavior. While DOC may shift marginal prescribing practices, mandatory PDMP policies and particularly Schedule V designation may induce more substantial shifts in treatment choice by increasing the effective relative price of gabapentin. Clinicians may respond by substituting toward unscheduled or less-regulated alternatives (e.g., physical therapy, over-the-counter formulations), or conversely, toward other Schedule V or Schedule IV medications that remain similarly accessible, such as tramadol, which has its own misuse potential (Gupta et al., 2023). Importantly, several opioids used in outpatient pain management, including tramadol (Schedule IV) and codeine combinations (Schedule V), may be seen as administratively more familiar to prescribers, despite their higher risks.

These trade-offs mirror findings from earlier upscheduling policies. For instance,

Gupta et al. (2023) find that upscheduling tramadol did not reduce total opioid prescribing due to substitution toward other opioids. Similarly, Grauer and Cramer (2022) show that gabapentin’s Schedule V designation is associated with a reduction days supplied in Medicare Part D, but downstream impacts on substitution nor mortality were not evaluated. The current evidence is therefore incomplete on whether stricter classification improves patient safety or merely shifts prescribing patterns in unintended ways.

2.3 Conceptual Overview

Regulatory restrictions on gabapentin prescribing may influence its prescription dispensation through three related channels: (1) *risk salience*, in which regulatory labeling may heighten providers’ awareness of misuse risk, leading them to prescribe more cautiously even without new information. (2) *information*, in which when combined with mandatory PDMP access, these policies reveal patient-specific controlled-substance histories, allowing prescribers to incorporate new information about potential misuse into their decisions. And, (3) *hassle costs*, where regulatory requirements such as documentation, refill limits, and mandatory PDMP checks impose administrative burdens that may deter (frequent) prescribing independent of clinical or informational considerations.

Together, these mechanisms provide a conceptual framework for understanding how state-level policy interventions can alter provider behavior through changes in perceived risk, available information, and the cost of compliance.

The next section develops a formal model that embeds these channels and derives empirical predictions summarized in Table 3.

3 Theoretical Framework

3.1 Provider Decision Problem

We model a provider’s decision to prescribe gabapentin to patient i as a tradeoff between expected therapeutic benefit, perceived risk of misuse, and the hassle cost imposed by administrative burdens of prescribing. Let the utility from prescribing be:

$$U_i = \alpha B_i - \lambda C(r_i) - h(s, \kappa),$$

where B_i denotes the expected clinical benefit to the patient, $C(r_i)$ is the perceived cost of prescribing that depends on the patient’s observed or inferred risk profile r_i , λ is

the weight the provider places on this perceived risk, and $h(s, \kappa)$ captures the hassle cost induced by administrative or regulatory burden of prescribing that depends on the state policy environment s and PDMP design features κ . The provider prescribes if $U_i \geq 0$, which defines an implicit threshold rule:

$$\tau(r_i; s, \kappa) = \frac{\lambda C(r_i) + h(s, \kappa)}{\alpha}.$$

A prescription is issued when $B_i \geq \tau(r_i; s, \kappa)$.

3.2 Impact of Stricter Prescribing Restrictions

Tighter controls on gabapentin prescribing—through (1) the Drug of Concern (DOC) designation and (2) Schedule V (C–V) classification—can shift prescribers’ decision calculus. Each channel operates through distinct mechanisms that affect the perceived risk λ , the information component $C(r_i)$, or the hassle cost of prescribing induced by administrative burdens of prescribing $h(s, \kappa)$.

1. Risk Salience.

The DOC or C–V designation increases the salience of potential misuse and regulatory scrutiny, thereby raising the prescriber’s perceived weight on risk, λ . In states without mandatory-access PDMPs, the DOC designation primarily enhances *salience* rather than providing additional information or imposing new hassle costs. Consequently, declines in gabapentin prescribing following DOC designation in the absence of PDMP mandates would most plausibly reflect a salience-driven response. The heightened salience increases the prescribing threshold for all patients, particularly those who are gabapentin-naïve, leading to larger reductions in initiation relative to continuation.

2. Information Effect.

In states with mandatory-access PDMPs, the DOC designation also enhances informational transparency by revealing patient-level prescribing histories, including co-prescribing of controlled substances. This makes individual risk signals, r_i , observable, allowing prescribers to update the perceived risk cost, $C(r_i)$, upward for high-risk patients. Formally, the policy generates an *information shock*:

$$C'(r_i) = C(r_i) + \Delta C(r_i), \quad \text{where } \Delta C(r_i) > 0 \text{ if } r_i \text{ indicates elevated misuse risk.}$$

This mechanism predicts larger prescribing reductions among patients flagged as high-risk—such as those with existing prescriptions of which their providers will now be aware.

3. Hassle Cost.

Schedule V (C-V) classification imposes additional administrative burdens, represented by $h(s, \kappa)$, such as mandatory PDMP queries, refill restrictions, and increased visit frequency. These fixed per-prescription costs likely discourage prescribing on the extensive margin (fewer treated patients) among both preexisting and new users.

4 Data and Methods

4.1 Policy Information

The dates of policy changes around gabapentin’s classification in state PDMPs differs between sources, including Grauer and Cramer (2022), Karavolis et al. (2021), and information from PDMP Assist, leading us to investigate each policy change through state laws and announcements. From 2017-2023, 7 states classified gabapentin as a Schedule V substance and 17 states and the District of Columbia classified it as a *drug of concern*. Figure 1 shows the geographic distribution of each policy across the US. The exact dates and sources for each policy change is shown in Table A1. Throughout our sample period, Nebraska required reporting of all prescription drugs in their PDMP, and thus will not be included in our analyses.

4.2 Health Insurance Claims

Our main measures of health-care use comes from the Merative MarketScan® Research Databases, a longitudinal panel of employer-sponsored insurance claims covering 2016–2023. MarketScan tracks employees and their covered dependents month-by-month, linking every paid claim or capitated encounter to eligibility files and basic demographics. The data span inpatient, outpatient, retail-pharmacy, and carve-out services and are supplied by roughly 350 large, self-insured employers. After applying our inclusion rules, the analytic sample contains about 75 million unique enrollees, including 53 million employee policy-holders. As shown in Table 2, an average state sees 414,262 enrollees per quarter. Gabapentin prescriptions were identified by collecting NDC codes

from MarketScan’s RED BOOK³ which had product names containing any of the following: “gabapentin”, “horizant”, “gralise”, “neurontin”, “gabarone”, or “fusepaq fanatrex”. As shown in Table 2, an average of 12 patients per 1,000 are prescribed gabapentin in each state per quarter. Pregabalin prescriptions were identified with the same method using product names containing “pregabalin” or “lyrica”. Opioid prescriptions were identified using their ATC classification code “N02A,” and mapping this to all corresponding NDC codes, which were then flagged as such in the data. Nonpharmacologic therapy for pain procedures were identified using CPT codes identified in Table 3 of Lee et al. (2021) in both the outpatient and inpatient claims files.

Using the same NDC codes to identify gabapentin and pregabalin, we extract the same estimates from the public use Medicaid State Drug Utilization Data (SDUD) files and Medicare Part D prescribing data. To calculate rates, we use state Medicaid and Medicare enrollee estimates from the Kaiser Family Foundation’s (KFF) State Health Facts tables “Health Insurance Coverage of the Total Population,” which provides estimates of individuals using each type of insurance from 2016-2024.

4.3 Mortality

Mortality data from 2010 to 2023 were obtained from the National Center for Health Statistics’ NVSS multiple cause-of-death files, which comprise the complete census of U.S. deaths. Overdose fatalities were identified via ICD-10 external cause-of-injury codes X40–X44, X60–X64, X85, and Y10–Y14. Anticonvulsant involvement—including gabapentinoids and Z-drugs—was flagged using codes T42.6 (“other antiepileptic and sedative-hypnotic drugs”) and T42.7 (“unspecified antiepileptic and sedative-hypnotic drugs”) following Powell et al. (2023). Opioid-related deaths were defined by codes T40.0–T40.4 and T40.6 in accordance with Ruhm (2018). Decedents with both an anticonvulsant and an opioid code were classified as combination-involved overdoses. Although T42.6 and T42.7 do not explicitly name gabapentinoids or Z-drugs, these agents are encompassed within those categories in the standard ICD-10 classification. We use the Epidemiology and End Results (SEER) population statistics are used to create mortality rates.

³“The RED BOOK Table is a supplemental table that provides additional information about prescription drugs (for example, generic name, manufacturer, therapeutic class). Drugs are listed in this file by National Drug Code. The code is linkable to the Prescription Drug Claims (D) Table, so that selection of drug claims may be made by the categorical fields included in the RED BOOK,” MarketScan User Guide

Figure A1 plots the number of gabapentinoid-related deaths over time. Although, the total number deaths are relatively small, it is clear that these deaths have been increasing over time. Since 2010, the number of gabapentinoid related deaths have increased ten fold, with almost 90-percent of these deaths involving an opioid by 2023. Of the roughly 81,000 opioid-related deaths in 2023, approximately 12-percent of them involved a gabapentinoid.

We also see a similar trend in Figure A2, which presents geographic trends in toxicology-confirmed gabapentin-involved poisonings reported to the fifty-five American Association of Poison Control Centers (AAPCC) and recorded in the National Poison Data System (NPDS) from 2011 to 2017.⁴ These maps visualize 3-digit ZIP code-level rates of Poison Help calls involving gabapentin exposures, adjusted for population and categorized using percentile cutoffs anchored to 2013 call volumes. Over this period, gabapentin-related poisoning rates increased in both incidence and geographic spread. This rise in non-fatal and fatal gabapentin-involved poisonings highlights an important public health concern, especially given that Poison Help call data reflect real-time toxic exposure reports and include cases beyond overdose deaths. As these confirmed poisoning episodes—over 60% of which originate from health care facilities—often serve as early indicators of misuse, they may be especially responsive to regulatory efforts like controlled substance scheduling or drug-of-concern designations. The observed trends reinforce the need to systematically evaluate how prescribing controls affect gabapentin use and misuse in both monotherapy and poly-substance contexts.

4.4 Empirical Strategy

We exploit the staggered adoption of two distinct state-level policy interventions—(i) Schedule V classification of gabapentin and (ii) “drug of concern” designation requiring PDMP monitoring—to estimate their causal effects on prescribing and downstream outcomes. Our primary identification strategy is a stacked difference-in-differences (DiD) event-study framework, which pools cohorts of states adopting the policy in different years while flexibly controlling for calendar and cohort trends (Cengiz et al., 2019; Wing et al., 2024). Let s index states, t calendar quarter-year, and g the “cohort” equal to the

⁴Unfortunately, we do not have access to these data past 2017, and the first year of Schedule V designation is 2017.

quarter-year in which state s first implements the relevant policy. We estimate

$$Y_{g,s,t} = \alpha_s + \zeta_t + \eta_{g,t} + \sum_{\substack{k=-8 \\ k \neq -1}}^8 \beta_k D_{g,s,t}^{(k)} + \varepsilon_{g,s,t}, \quad (1)$$

where $Y_{g,s,t}$ is the outcome of interest (e.g., gabapentin prescriptions per 1,000 population, opioid co-prescribing rates, non-pharmacologic therapy visits, or overdose mortality), α_s are state-by-cohort fixed effects and $\eta_{g,t}$ are cohort-by-quarter-year fixed effects. The indicator $D_{g,s,t}^{(k)}$ is equal to one when state s is in event-quarter-year k relative to its adoption quarter-year g (with $k = -1$ omitted), and zero otherwise. Standard errors are clustered at the state level to allow for arbitrary serial correlation (Bertrand et al., 2004). Coefficients β_k trace the dynamic response before and after adoption; small and insignificant estimates of β_k for $k < 0$ supports the parallel-trends assumption, while β_k for $k \geq 0$ capture the timing and persistence of treatment effects.

For ease of interpretation, we also estimate a collapsed stacked DiD model:

$$Y_{s,t} = \alpha_s + \zeta_t + \delta \text{Treat}_{s,t} + \mathbf{X}_{s,t} \psi + u_{s,t}, \quad (2)$$

where $\text{Treat}_{s,t}$ equals to one for all $t \geq g$ in treated state s , and zero otherwise. Here, δ measures the average post-policy change across treated states.

4.5 Robustness Checks and Empirical Tests of Mechanisms

After establishing the overall impact of tighter prescribing restrictions on gabapentin use, we next examine the mechanisms through which these policies operate. The theoretical framework in Section 3 suggests three primary channels: (i) increased perceived risk of gabapentin misuse (*salience*), (ii) additional patient-specific information revealed through PDMP queries (*information*), and (iii) increased administrative burden associated with tighter prescribing controls (*hassle cost*). We conduct a series of robustness and heterogeneity tests to evaluate the three channels—salience, information revelation, and hassle cost—that may explain the observed relationship between tighter regulation of gabapentin and changes in its’ prescription dispensing. Each test exploits variation across policy environments, prescriber types, and patient subgroups.

Salience Effect (Risk Perception Channel).

A decline in prescribing following DOC adoption, even in states without access to new

patient-specific information (via MA-PDMPs) or additional administrative requirements (via Schedule V reclassification), would indicate that providers respond to increased salience of perceived risk associated with gabapentin prescribing. We test this hypothesis using a difference-in-differences (DiD) model comparing prescribing and harm outcomes in states that labeled gabapentin as a DOC but did not operate a mandatory-access PDMP ($MA-PDMP = 0$) to those in never-treated states:

$$Y_{ist} = \alpha_i + \delta_s + \theta_t + \beta^{SAL} DOC_{st} + \varepsilon_{ist}, \quad (3)$$

where Y_{ist} denotes a prescribing outcome (e.g., the probability that patient i in state s and quarter t receives any gabapentin). A negative and statistically significant β^{SAL} would be consistent with a salience-driven reduction in prescribing, operating through heightened provider perception of misuse risk rather than through information revelation or administrative frictions.

Because initiation decisions are more discretionary than continuation of existing therapy, we expect heightened salience of gabapentin-related risks to have a larger impact on new starts rather than ongoing use. We test this by comparing effects for gabapentin naive vs existing gabapentin users by stratifying this model by each type. A larger β^{SAL} for the new users would be consistent with a stronger salience effect on initiation than continuation.

Information Effect (Patient-Specific Signal Channel).

When DOC designation coincides with a mandatory-access PDMP, providers obtain new patient-specific information on prior controlled-substance use. If this information reveals greater patient-specific misuse risk, prescribing should decline more sharply in MA-PDMP states, especially for patients with a documented history of high-risk use. We test this by comparing changes in prescribing and harm outcomes between states that designate gabapentin as a DOC with and without MA-PDMP requirements, estimating:

$$\begin{aligned} Y_{ist} = & \alpha_i + \delta_s + \theta_t + \beta_1^{INF} DOC_{st} \\ & + \beta_2^{INF} (DOC_{st} \times MA-PDMP_s) \\ & + \beta_3^{INF} (DOC_{st} \times MA-PDMP_s \times Existing_i) + \varepsilon_{ist}. \end{aligned} \quad (4)$$

where $Existing_i$ indicates patients with prior exposure to gabapentin. A negative β_2^{INF} would suggest that the DOC effect is amplified in MA-PDMP states, while $\beta_3^{INF} <$

0 would confirm larger declines among existing patients, consistent with providers incorporating newly revealed patient-level risk information into prescribing decisions.

Hassle-Cost Effect (Administrative Burden Channel).

Finally, we test whether the additional administrative requirements associated with Schedule V (C–V) reclassification—such as refill limits and PDMP checks—increase the per-prescription hassle cost. We estimate:

$$Y_{ist} = \alpha_i + \delta_s + \theta_t + \beta^{HAS} CV_{st} + \varepsilon_{ist}, \quad (5)$$

where as above, Y_{ist} denotes a prescribing outcome. A negative and statistically significant β^{HAS} would indicate the hassle cost driving a reduction in prescribing through administrative frictions, on top of the salience and information still present.

We will again test the effects on continuation versus initiation by stratifying this model by gabapentin naive and existing gabapentin users. A larger β^{HAS} for the new users would be consistent with a stronger hassle cost effect on initiation.

Together, these tests provide evidence on the whether and how the three mechanisms - (i) risk-salience shifts in λ , (ii) information-driven updates to perceived misuse risk $C(r_i)$, and (iii) administrative frictions reflected in $h(s, \kappa)$ - may contribute to the changes in gabapentin prescription dispensing following tighter prescribing regulations. Each regression complements the unified event-study and DiD models and provides mechanism-specific validation of the theoretical framework.

4.6 Descriptive Analysis

The relationship between gabapentin drug monitoring gabapentin prescribing is presented in Figure 2, which plots the number of patients with a gabapentin prescription (per 1,000) relative to 2016 over time for both drugs of concern states, Schedule V states, and those without a restriction. Initially, there is a slight decrease for Schedule V states, while drug of concern states and states without any gabapentin policy are on a similar trend. In 2019, there was a clear precipitous drop in prescribing in Schedule V states, which coincided with four of the seven states with such designation adopting this relatively stricter policy that year and all seven states having the designation by the end of this year. Conversely, drug of concern states seem to be on a similar trend throughout the sample period. Taken together, the trends show that gabapentin prescribing de-

creased relatively faster among Schedule V states. However, these results do not exploit variations in the timing of either policy. In the following sections, we run additional tests and examine the event-study estimates to provide more conclusive evidence of the effects of Schedule V and drug of concern policies.

5 Results

5.1 Effects of Gabapentin Regulation on Gabapentin Prescribing Behavior

Figures 3a and 3b present difference-in-differences event study estimates of the effects of state-level gabapentin regulation on prescribing. The outcome of interest is the share of patients with a gabapentin prescription, estimated in event time relative to the state-specific quarter of policy adoption (event time zero), with leads and lags centered around the quarter of implementation.

Figures 3a shows the impact of classifying gabapentin as a drug of concern, which triggers mandatory tracking in state Prescription Drug Monitoring Programs (PDMPs) but does not formally restrict prescribing. We observe no statistically or substantively meaningful change in gabapentin prescribing following this PDMP-only classification. Pre-treatment estimates are flat and centered around zero, supporting the validity of the parallel trends assumption. The post-treatment estimates are close to zero, and the confidence intervals overlap throughout, indicating that providers showed little to no change in behavior. In this analysis, we maintain Schedule V states in the control group. Similarly, when examining the treatment of Schedule V adoption we keep drugs of concern in the control group. The latter seems particularly appropriate given that we seen no difference between drugs of concern states and control states in terms of gabapentin prescribing. Nonetheless, Figures A3 and A4 show that the prescribing results are not appreciably different if we exclude these states within their respective analysis.

In contrast, Figure 3b shows the effects of Schedule V controlled substance designation, which imposes more stringent prescribing rules. Here, we observe a sustained and statistically significant decline in the share of patients with a gabapentin prescription beginning in the first quarter after policy implementation. The magnitude of the decline grows over time, reaching approximately 0.2 to 0.3 percentage points (on a baseline share of 3–4%) by 6–7 quarters post-policy. These effects are consistent across all three es-

timators and suggest that controlled substance scheduling, unlike PDMP-only tracking, meaningfully reduces gabapentin prescribing.

As these results rely on private commercial insurance claims, we use aggregated state level data to estimate the same outcomes in Medicaid and Medicare prescribing. Figure 4a and Figure 4b show similar decreasing impacts of Schedule V classification on prescriptions per 1,000 enrollees. Following the prior results, the drug of concern classification did not show significant changes for these payers as well, shown in Appendix Figures A5a and A5b. The results in Medicare Part D prescribing aligns with prior literature studying the association of the policies with Medicare prescribing (Grauer and Cramer, 2022).

Other policies during this time period may have impacted prescribing of pain medications, notably must access opioid PDMP policies and medical or recreational marijuana. We control for opioid MA-PDMP policies, legalization of medical marijuana, legalization of recreational marijuana, and the beginning of sales for recreational marijuana. None of these policies altered the main negative and significant impact of decreasing gabapentin prescriptions following the Schedule V classification, as shown in Figure 5.

Together, these figures demonstrate that stricter regulatory designation via Schedule V classification has a more pronounced effect on prescribing behavior than monitoring alone. These findings highlight the importance of regulatory intensity in shaping provider responses to prescription drug policy.

5.2 Impact on Prescribing of other Prescription Drugs with Misuse Potential

Reclassifying gabapentin as a Schedule V controlled substance brought new hurdles for prescribers, like limits on refills, extra paperwork, and requirements for electronic prescribing. These prescribing “costs” align gabapentin with other Schedule V substances, such as pregabalin, and with lower-schedule opioids (e.g., tramadol, codeine combinations), which may also be used for overlapping indications such as neuropathic or musculoskeletal pain. As a result, prescribers may be incentivized to shift away from gabapentin toward pharmacologic substitutes that now face similar administrative hurdles but may be perceived as more familiar, more potent, or more effective. To assess whether such substitution occurs, we examine trends in prescribing of pregabalin, opioids, and co-prescriptions involving gabapentin and opioids following Schedule V designation.

Figures 6 and 7 examine the potential substitution behaviors in response to the

Schedule V designation of gabapentin, focusing on changes in opioids (broadly defined), co-prescriptions of gabapentin with opioids, and prescribing of pregabalin (a pharmacologic analog). Estimates are again generated using a stacked difference-in-differences event study approach under staggered treatment timing.

Figure 6a investigates substitution toward prescription opioids more broadly. Here, we find no significant increase in opioid prescribing following gabapentin scheduling. Point estimates remain close to zero and statistically indistinguishable from pre-trend values, providing no clear evidence of net substitution toward opioids. Figure 6b assesses changes in the rate of co-prescribing of gabapentin and opioids. Consistent with the overall decline in gabapentin prescribing observed earlier, we document a statistically significant and persistent reduction in co-prescriptions beginning in the first quarter post-policy. These results suggest that Schedule V classification effectively reduces high-risk combinations of gabapentin and opioids, but only through a reduction in gabapentin prescribing.

Figure 7 shows a modest but statistically detectable increase in the share of patients receiving pregabalin prescriptions beginning roughly three quarters after the policy is implemented, suggesting that some prescribers may substitute toward pregabalin, another Schedule V anticonvulsant, when gabapentin prescribing is restricted.

Taken together, these findings imply that Schedule V designation of gabapentin may reduce risky co-prescribing without inducing substitution toward more harmful alternatives like opioids. These results are robust to controlling for other pain related policies, shown in Appendix Figure A10. However, the modest rise in pregabalin use highlights the possibility of therapeutic substitution toward other gabapentinoids, which may carry similar risks when co-used with opioids.

5.3 Impact on Gabapentin Involved Fatal and Non-fatal Adverse Events

In Figure 8, we present stacked difference-in-differences event study estimates of the impact of the Schedule V reclassification of gabapentin on mortality involving gabapentinoids. Panel (a) displays estimates for all deaths involving a gabapentinoid, while panel (b) narrows the outcome to those deaths involving both a gabapentinoid and an opioid.

Across both panels, we observe a notable post-controlled substance designation decline in mortality rates, with a relatively stable pre-trend and a clear downward shift in the quarters following the Schedule V classification (denoted by the vertical red dashed

line). This pattern suggests a substantial reduction in fatal adverse events temporally aligned with the regulatory intervention.

Specifically, the average treatment effect on the treated (ATT) is -0.109 (panel a) and -0.106 (panel b), both statistically significant at the 5 percent level. These values correspond to approximately a 10 percent reduction in gabapentin-involved mortality rates, measured per 100,000 deaths, relative to the pre-policy mean. The effect is particularly salient in panel (b), which reflects the high-risk subset of cases involving concurrent opioid use, consistent with clinical concerns about the potential of gabapentin to increase opioid toxicity.

The attenuation in mortality is gradual but persistent, suggesting that the Schedule V designation may have had a sustained deterrent effect on risky polypharmacy use or prescribing behavior. While confidence intervals remain relatively wide, especially at the end of the event-time window, the consistent negative trajectory suggests a meaningful decline attributable to tighter prescribing restrictions on gabapentin through its controlled substance designation.

In contrast, Appendix Figure A11 reports estimates from analogous event studies for nonfatal adverse events involving gabapentinoids following the classification of Schedule V. In contrast to significant reductions in mortality, the ATT estimate for nonfatal events is precisely estimated at zero ($ATT = < 0.000$), and no discernible post-policy trend. This null result suggests that while Schedule V classification may be effective in mitigating the most severe outcomes, such as fatal overdoses, it may have limited impact on the broader set of less severe adverse events or emergency department encounters.

Further comparison with Appendix Figures A8–A10, which present analogous estimates for states that designated gabapentin as a DOC, a weaker regulatory action, reinforces the importance of regulatory intensity. Panel (a) of Figure A8 shows no meaningful reduction in deaths involving gabapentinoids, and Figure A9 similarly shows no decline in gabapentinoid + opioid-involved fatalities. In fact, estimated ATTs in both cases are small and statistically insignificant ($ATT = 0.037$ and 0.057 , respectively). Figure A10 further confirms a lack of meaningful effect of DOC designation on non-fatal adverse events, with a near-zero ATT estimate (0.030) and wide confidence intervals throughout the post-policy period. These Schedule V classification results are also robust to controlling for other pain related policies, as shown in Appendix Figure A15.

The contrast between Figures 6 and A8–A10 underscores the importance of the regulatory intensity of policy interventions. While DOC classifications may serve as

informational tools, our findings suggest that they are insufficient to meaningfully alter fatal or non-fatal adverse event trajectories. In contrast, the Schedule V designation—accompanied by stricter prescribing and dispensing controls—appears to exert a measurable and sustained impact on high-risk use and mortality.

Together, these findings support the hypothesis that regulatory scheduling of gabapentin may be effective in mitigating its most severe public health risks, particularly when co-used with opioids. This is consistent with broader efforts to constrain the supply-side drivers of polypharmacy-related overdose deaths.

5.4 Impacts by Prior Gabapentin Exposure

To examine how effects differ for patients initiating versus continuing gabapentin therapy, we stratify the continuously-enrolled MarketScan sample (2,851,760 individuals enrolled in all quarters from 2016 through 2023) into three cohorts based on prescription activity in the baseline year, 2016:

- **Gabapentin users:** individuals with at least one gabapentin prescription in 2016 (68,525)
- **Gabapentin naive:** individuals with no gabapentin prescription in 2016 (2,783,235)
- **Broad naive:** individuals with no prescriptions for gabapentin, pregabalin, or opioids in 2016 (2,008,185)

The gabapentin naive cohort includes all individuals in the broad naive group, but also those who may have used pregabalin or opioids in 2016. The broad naive group represents individuals with no prior exposure to either gabapentinoids or opioids and therefore provides a clean sample of new potential entrants into medications with these restrictions.

We estimate the specification in Equation 2 separately by group to examine prescribing of gabapentin, pregabalin, opioids, and other mental-health medications, as well as emergency department (ED) visits. Mortality cannot be observed in MarketScan, so ED visits serve as our measure of severe adverse outcomes.

Figure 9 presents the results. All groups experienced large and statistically significant declines in gabapentin prescribing, with the largest effects among baseline gabapentin users, indicating substantial reductions in continuation. Reductions are also evident

among naive groups, demonstrating reduced initiation into gabapentin use. Consistent with a hassle-cost mechanism, pregabalin prescribing increases significantly for the gabapentin-naive and broad-naive cohorts, indicating therapeutic substitution when both medications become equivalently regulated. We also find increases in other mental-health drug prescribing among the gabapentin-naive cohort, consistent with substitution for off-label anti-anxiety and antidepressant uses of gabapentin. Opioid prescribing declines in the naive cohorts as well, suggesting spillovers into pain management.

ED visits decline across all cohorts, suggesting that increased gabapentin controls do not worsen acute clinical outcomes and may reduce severe events among baseline users. Taken together, these results indicate meaningful reductions in both continuation and initiation of gabapentin therapy, substitution toward other analgesic and psychiatric medications, and no evidence of increased acute harms.

5.5 Empirical Tests of Mechanisms

To empirically test the mechanisms described in Section 4.5, we estimated the equations to see how each additional layer of policy impacted (1) Risk Salience, (2) Information, and (3) Hassle Cost. We use the continuously enrolled cohort from the previous section and look at results for the full sample, existing gabapentin users, and gabapentin naive.

First, to understand how salience alone may impact prescribing behavior, we look to states who implemented the drug of concern classification but did not have a mandatory-access PDMP (MA-PDMP). As shown in columns (1)-(3) of Table 4, there is no impact of these policies for the overall sample nor for either existing gabapentin or naive patients. This indicates that salience alone is not affective in altering prescribing behavior, as was the case for opioid PDMP studies [CITE].

To understand how additional information from accessing the PDMP in must-access states altered prescribing behavior, we look at all cohorts and estimate the impact of the drug of concern policy, specifically in must-access states, and for existing users, who may be at higher risk. These results are presented in column (4) of Table 4. The policy on its own in a must-access state or not does not significantly alter prescribing. However, there is a large and significant decrease for existing users, which provides evidence that prescribers incorporate the information in the PDMP for their patients, particularly those who have a record for prior proscriptions.

Finally, we want to understand how the strongest policy of a Schedule V classification

impacts prescribing through hassle cost. We estimate the impact on the share of patients with a prescription among the full continuously enrolled sample, the existing users, and the gabapentin-naïve. As shown in column (1) of Table 5, there is a negative and significant effect on prescribing in the full sample. This is driven by the large decrease seen for existing users, while there is no significant impact for new users.

6 Discussion and Conclusion

This study provides novel evidence on the prescribing consequences of increased regulation of gabapentin, a drug that has seen substantial off-label growth in recent years and has become increasingly associated with misuse, particularly in combination with opioids. Leveraging the staggered implementation of state-level policies classifying gabapentin as either a drug of concern (DOC) or as a Schedule V controlled substance, we examine how prescribers respond to regulatory friction of varying intensity.

Our findings reveal that classifying gabapentin as a DOC, which entails inclusion in PDMPs but does not restrict prescribing, had little to no impact on prescribing rates. In contrast, Schedule V classification, which introduces stricter controls and documentation requirements, resulted in a sustained and statistically significant reduction in the share of patients receiving gabapentin prescriptions. These patterns suggest that regulatory designations must impose real prescribing costs to generate meaningful behavioral change, a result broadly consistent with prior work examining the rescheduling of opioids such as hydrocodone and tramadol (Gupta et al., 2023).

In assessing potential unintended consequences, we find limited evidence of clinically concerning substitution behavior. There is no detectable increase in prescribing of opioids, and co-prescribing of gabapentin and opioids declines following scheduling. However, we do observe a modest increase in pregabalin prescribing, a pharmacologic analog of gabapentin that was already Schedule V. This is consistent with economic models of substitution, in which prescribers shift to alternative treatments facing similar regulatory barriers when relative prescribing costs are equalized. While such substitution may not carry the same overdose risk as opioid substitution, it still merits clinical scrutiny given emerging evidence on the misuse potential of gabapentinoids.

Our findings contribute to a growing literature evaluating supply-side drug policy interventions in settings where therapeutic and abuse-prone substitutes exist. They underscore the importance of considering the behavioral response of prescribers when

designing regulatory interventions, especially for drugs that occupy ambiguous clinical and regulatory space. From a policy perspective, our results suggest that regulatory classifications short of scheduling may be insufficient to curb use in high-risk contexts, and that policies must weigh potential substitution channels even when targeting non-opioid therapies.

Future work should assess patient-level outcomes, such as pain management effectiveness, adverse events, and healthcare utilization, to more fully characterize the clinical trade-offs of gabapentin scheduling. More broadly, as regulators continue to evaluate gabapentinoids and other off-label therapies, our study provides a framework for anticipating how prescribers may respond to differential policy levers and the resulting implications for public health.

References

- Alpert, A., Dykstra, S., and Jacobson, M. (2024). Hassle costs versus information: How do prescription drug monitoring programs reduce opioid prescribing? *American Economic Journal: Economic Policy*, 16(1):87–123.
- Alpert, A., Powell, D., and Pacula, R. L. (2018). Supply-side drug policy in the presence of substitutes: Evidence from the introduction of abuse-deterrent opioids. *American Economic Journal: Economic Policy*, 10(4):1–35.
- Bao, Y., Pan, H., Taylor, H. A., Radakrishnan, B. R., and Barry, C. L. (2016). Prescription drug monitoring programs are associated with sustained reductions in opioid prescribing by physicians. *Health Affairs*, 35(6):1045–1051.
- Bertrand, M., Duflo, E., and Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119(1):249–275.
- Buchmueller, T. C. and Carey, C. (2018). The effect of prescription drug monitoring programs on opioid utilization in medicare. *American Economic Journal: Economic Policy*, 10(1):77–112.
- Bykov, K., Bateman, B. T., Clark, K., et al. (2020). Association of gabapentinoids with the risk of opioid-related adverse events in surgical patients in the united states. *JAMA Network Open*, 3(12):e2031647.
- Cawley, J. and Rizzo, J. (2008). Spillover effects of prescription drug withdrawals. *Advances in Health Economics and Health Services Research*, 19:119–144.
- Cengiz, D., Dube, A., Lindner, A., and Zipperer, B. (2019). The effect of minimum wages on low-wage jobs. *The Quarterly Journal of Economics*, 134(3):1405–1454.
- Collins, J. M., Simon, K. I., and Tennyson, S. (2013). Drug withdrawals and the utilization of therapeutic substitutes: The case of viox. *Journal of Economic Behavior & Organization*, 86:148–168.
- Drug Enforcement Administration (2020). Drugs of Abuse.
- Finkelstein, A., Hendren, N., and Shepard, M. (2019). Subsidizing health insurance for low-income adults: Evidence from massachusetts. *American Economic Review*, 109(4):1530–1567.
- Freedman, S., Kearney, M., and Lederman, M. (2012). Product recalls, imperfect information, and spillover effects: Lessons from the consumer response to the 2007 toy recalls. *Review of Economics and Statistics*, 94(2):499–516.
- Grauer, J. S. and Cramer, J. D. (2022). Association of state-imposed restrictions on gabapentin with changes in prescribing in medicare. *Journal of General Internal Medicine*, 37(14):3630–3637.

- Gupta, S., Nguyen, T., Freeman, P. R., and Simon, K. (2023). Competitive effects of federal and state opioid restrictions: Evidence from the controlled substance laws. *Journal of Health Economics*, 91:102772.
- Hah, J., Mackey, S. C., Schmidt, P., McCue, R., Humphreys, K., Trafton, J., Efron, B., Clay, D., Sharifzadeh, Y., Ruchelli, G., et al. (2018). Effect of perioperative gabapentin on postoperative pain resolution and opioid cessation in a mixed surgical cohort: a randomized clinical trial. *JAMA surgery*, 153(4):303–312.
- Karavolis, Z. A., Su, A. B., and Peckham, A. M. (2021). State-level response to gabapentin misuse in the united states: Implications and future direction. *American Journal of Health-System Pharmacy*, 79(9):e143–e148.
- Lavin, R. A., Liu, C., Yuspeh, L., Kalia, N., Leung, N., Tsourmas, N. F., Williams, L., Hunt, D. L., Bernacki, E. J., and Tao, X. G. (2021). Trends in the utilization and dose of gabapentinoids in combination with opioids in an injured worker population between 2008 and 2018. *Journal of occupational and environmental medicine*, 63(10):e694–e700.
- Lee, B., Yang, K.-C., Kaminski, P., Peng, S., Odabas, M., Gupta, S., Green, H. D. J., Ahn, Y.-Y., and Perry, B. L. (2021). Substitution of nonpharmacologic therapy with opioid prescribing for pain during the covid-19 pandemic. *JAMA Network Open*, 4(12):e2138453.
- Mattson, C. L., Chowdhury, F., and Gilson, T. P. (2022). Notes from the field: Trends in gabapentin detection and involvement in drug overdose deaths — 23 states and the district of columbia, 2019–2020. *Morbidity and Mortality Weekly Report (MMWR)*, 71(19):664–666.
- Meara, E., Horwitz, J. R., Powell, W., McClelland, L., Zhou, W., O’Malley, A. J., and Morden, N. E. (2016). State legal restrictions and prescription-opioid use among disabled adults. *New England Journal of Medicine*, 375(1):44–53.
- Modesto-Lowe, V., Barron, G. C., Aronow, B., and Chaplin, M. (2019). Gabapentin for alcohol use disorder: A good option, or cause for concern? *Cleveland Clinic Journal of Medicine*, 86(12):815–823.
- Parkinson, K., Price, J., Simon, K. I., and Tennyson, S. (2014). The influence of fda advisory information and black box warnings on individual use of prescription antidepressants. *Review of Economics of the Household*, 12(4):771–790.
- Pauly, N. J., Delcher, C., Slavova, S., Lindahl, E., Talbert, J., and Freeman, P. R. (2020). Trends in gabapentin prescribing in a commercially insured us adult population, 2009–2016. *Journal of managed care & specialty pharmacy*, 26(3):246–252.
- Peckham, A. M., Evoy, K. E., Ochs, L., and Covvey, J. R. (2018). Gabapentin for off-label use: Evidence-based or cause for concern? *Substance Abuse: Research and Treatment*, 12:1178221818801311.

- Powell, D., Peet, E. D., and Pacula, R. L. (2024). Understanding the rise in overdose deaths involving opioids and non-opioid prescription drugs in the united states. *International Journal of Drug Policy*, 104:104104.
- Powell, D., Shetty, K. D., and Peet, E. D. (2023). Trends in overdose deaths involving gabapentinoids and z-drugs in the united states. *Drug and alcohol dependence*, 249:109952.
- Ruhm, C. J. (2018). Corrected us opioid-involved drug poisoning deaths and mortality rates, 1999–2015. *Addiction*, 113(7):1339–1344.
- Sacks, D. W., Hollingsworth, A., Nguyen, T., and Simon, K. (2021). Can policy affect initiation of addictive substance use? evidence from opioid prescribing. *Journal of Health Economics*, 76:102397.
- Smith, R. V., Havens, J. R., and Walsh, S. L. (2016). Gabapentin misuse, abuse and diversion: A systematic review. *Addiction*, 111(7):1160–1174.
- Torrance, N., Veluchamy, A., Zhou, Y., Fletcher, E. H., Moir, E., Hebert, H. L., Donnan, P. T., Watson, J., Colvin, L. A., and Smith, B. H. (2020). Trends in gabapentinoid prescribing, co-prescribing of opioids and benzodiazepines, and associated deaths in scotland. *British journal of anaesthesia*, 125(2):159–167.
- U.S. Food and Drug Administration (2019). Fda warns about serious breathing problems with seizure and nerve pain medicines gabapentin (neurontin, gralise, horizant) and pregabalin (lyrica, lyrica cr). <https://www.fda.gov/drugs/drug-safety-and-availability/fda-warns-about-serious-breathing-problems-seizure-and-nerve-pain-medicines-gabapentin-neurontin>.
- Wing, C., Freedman, S. M., and Hollingsworth, A. (2024). Stacked difference-in-differences. Technical report, National Bureau of Economic Research.

Figures

Figure 1: Policy Variation Across States

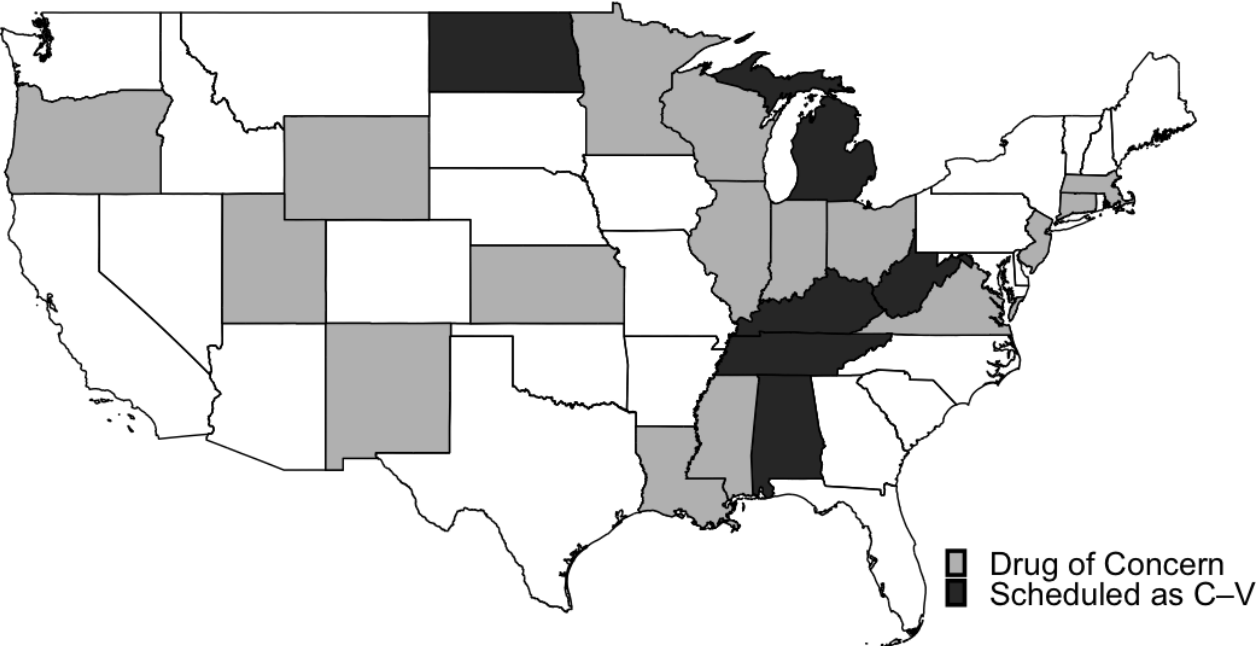
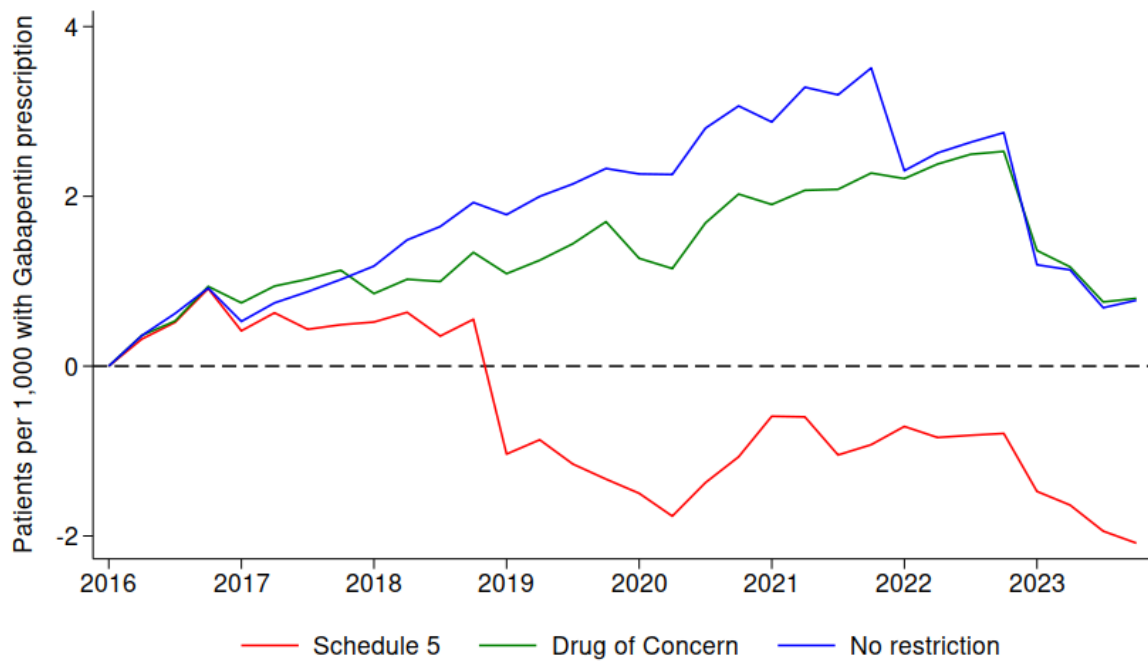


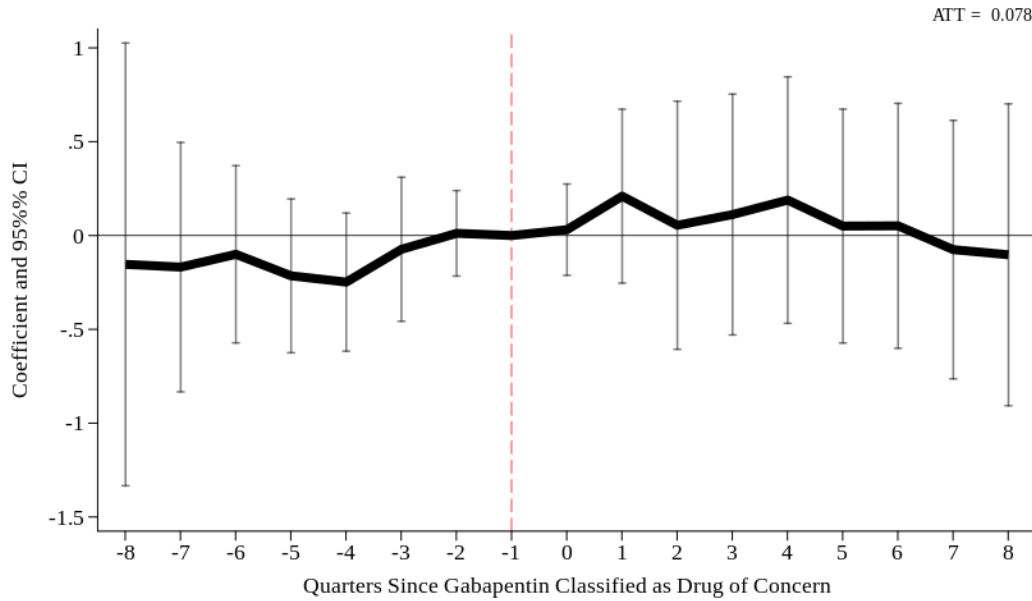
Figure 2: Gabapentin Prescriptions by State Restriction Policy



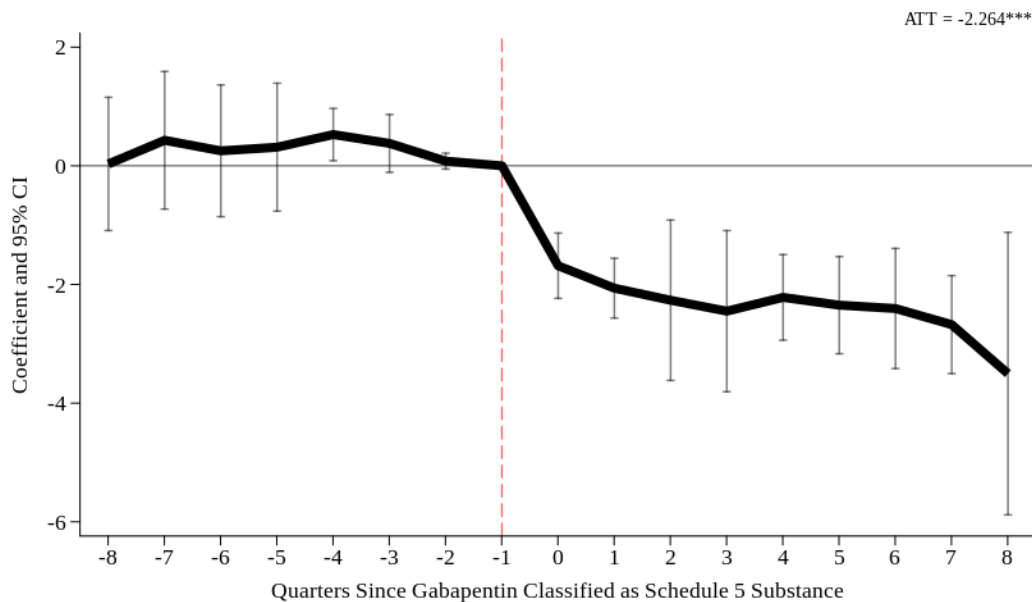
Gabapentin prescriptions per 1,000 patients by state policy of ever classifying Gabapentin as a Schedule V controlled substance, ever as a drug of concern, or no restriction through the PDMP. Rates are normalized to 2016, the first year of our sample period.

Figure 3: Gabapentin: Drugs of Concern vs. Schedule V Classification

(a) Drug of Concern



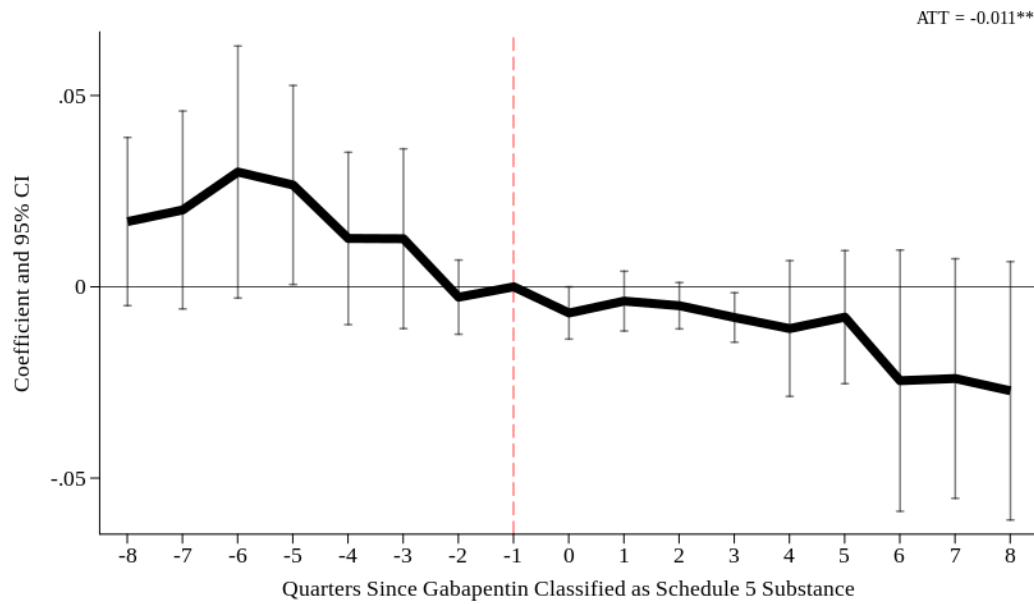
(b) Schedule V



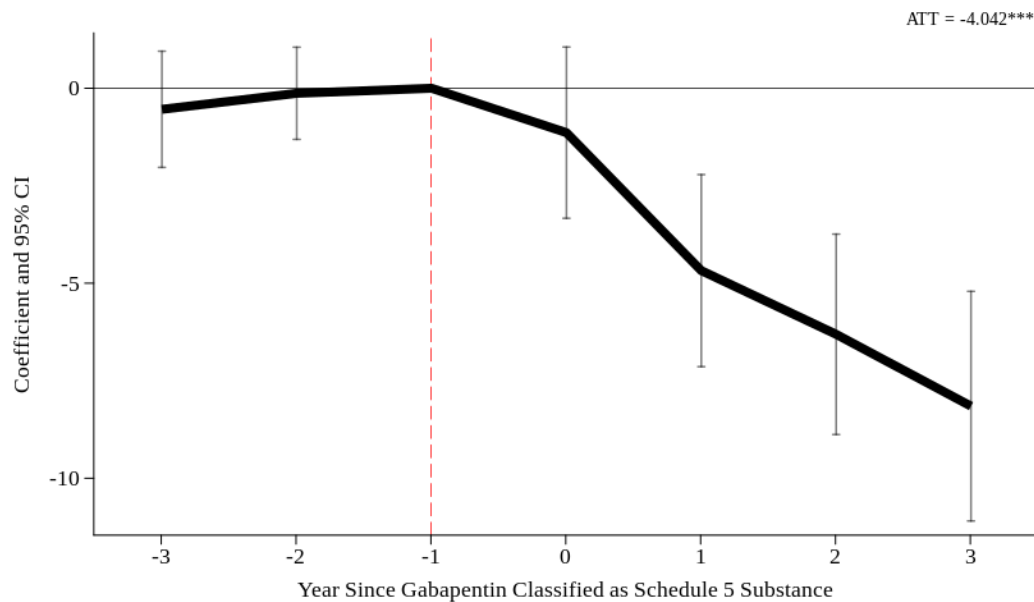
Note: (a) Change in rate of patients per 1,000 with a Gabapentin prescription following classification in state to drug of concern, and thus tracked in the PDMP. (b) Change in rate of patients per 1,000 with a Gabapentin prescription following classification in state to Schedule V controlled substance.

Figure 4: Gabapentin: Schedule V Classification - Public Payer

(a) Medicaid

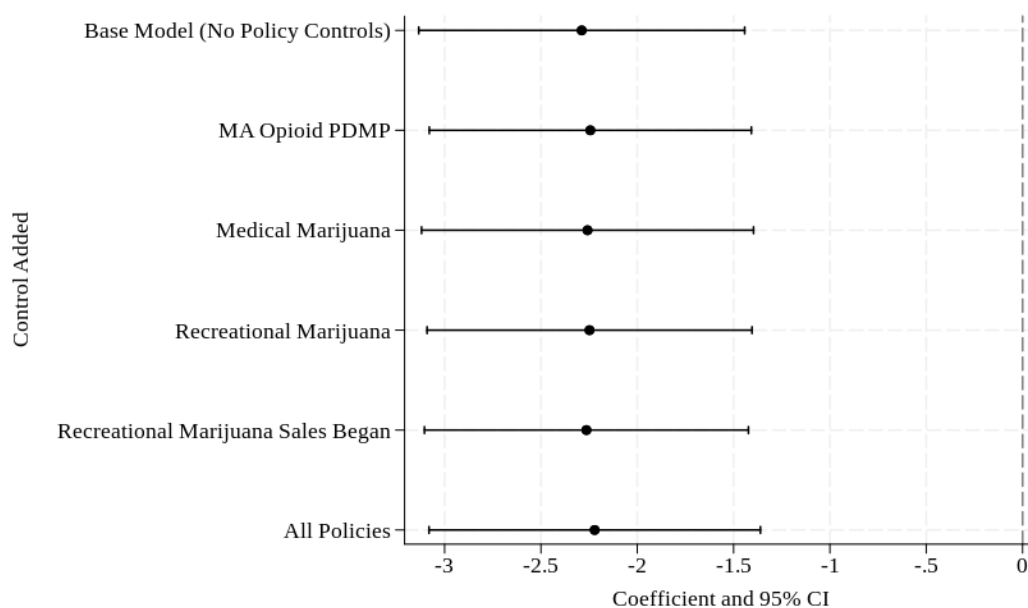


(b) Medicare



Note: (a) Change in rate of rate of Medicaid enrollees per 1,000 with a Gabapentin prescription following classification in state to Schedule V controlled substance. (b) Change in rate of rate of Medicare Part D enrollees per 1,000 for an average quarter each year with a Gabapentin prescription following classification in state to Schedule V controlled substance.

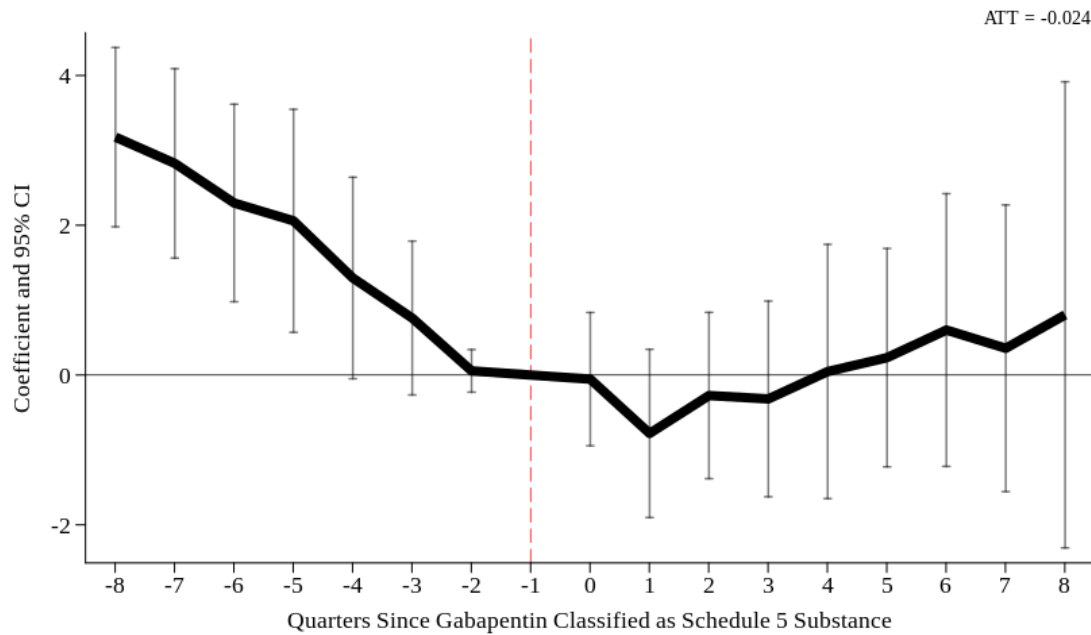
Figure 5: Gabapentin: Schedule V Classification, Other Policies Robustness Check



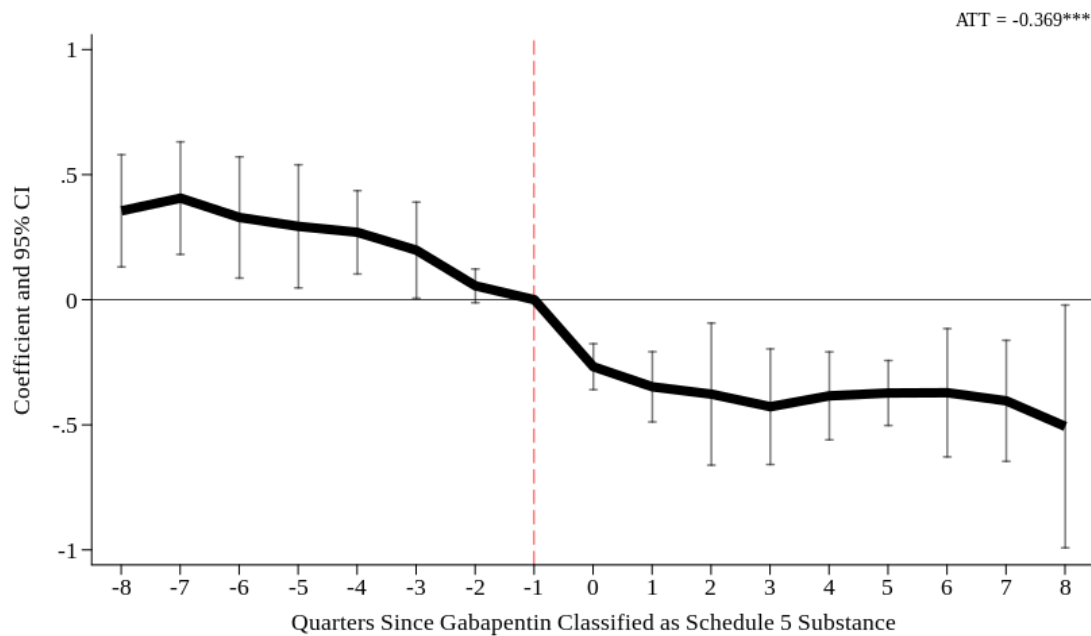
Note: Change in rate of patients per 1,000 with a Gabapentin prescription following classification in state to Schedule V controlled substance. Results controlling for must access PDMP and recreational or medical marijuana policies are shown with each corresponding point estimate, with the bottom result controlling for all policies.

Figure 6: Schedule V Classification Effects on Opioid Prescribing

(a) Opioids

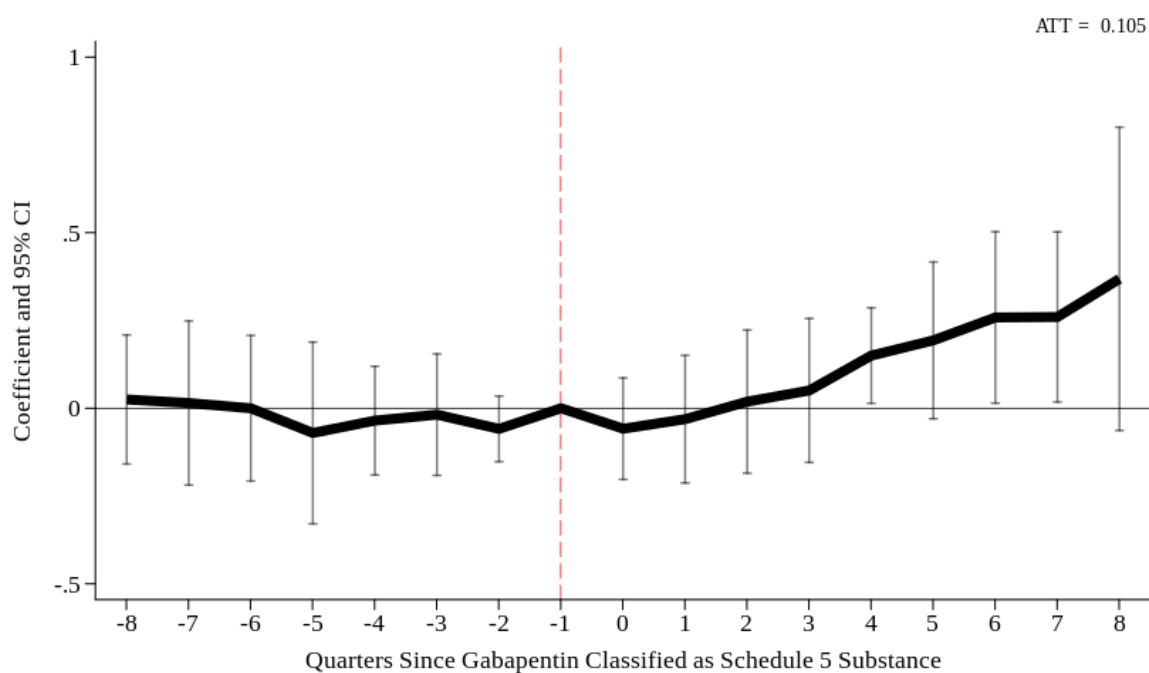


(b) Co-prescribing Opioids and Gabapentin



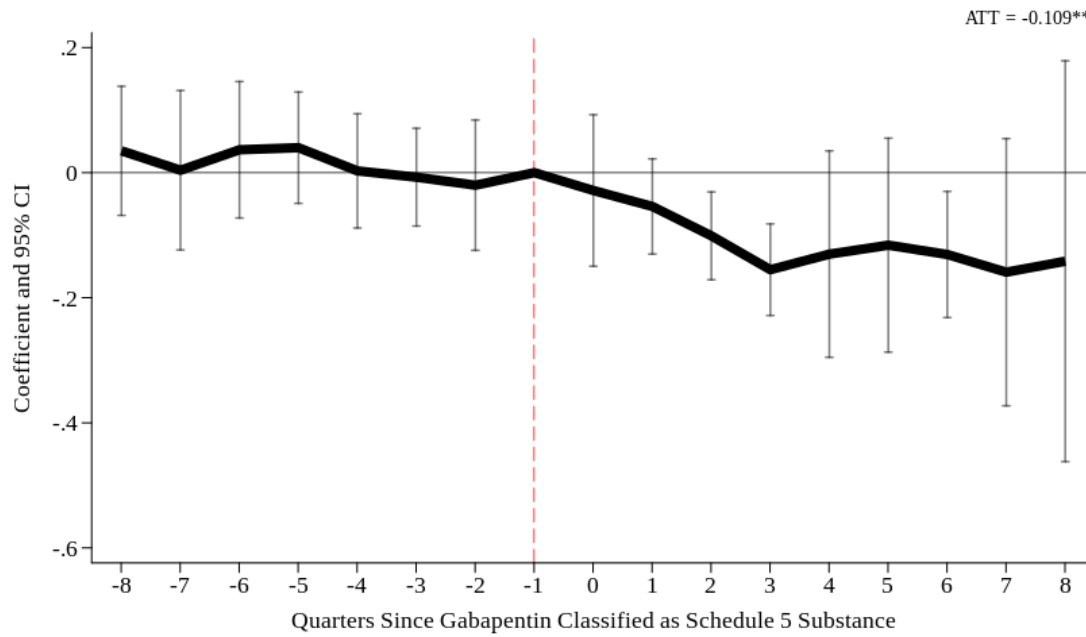
Note: (a) Change in rate of patients per 1 000 with an opioid prescription following classification of Gabapentin in state to Schedule V controlled substance. (b) Change in rate of patients per 1 000 co-prescribed Gabapentin and an opioid following classification of Gabapentin in state to Schedule V controlled substance.

Figure 7: Schedule V Classification - Pregabalin Impact

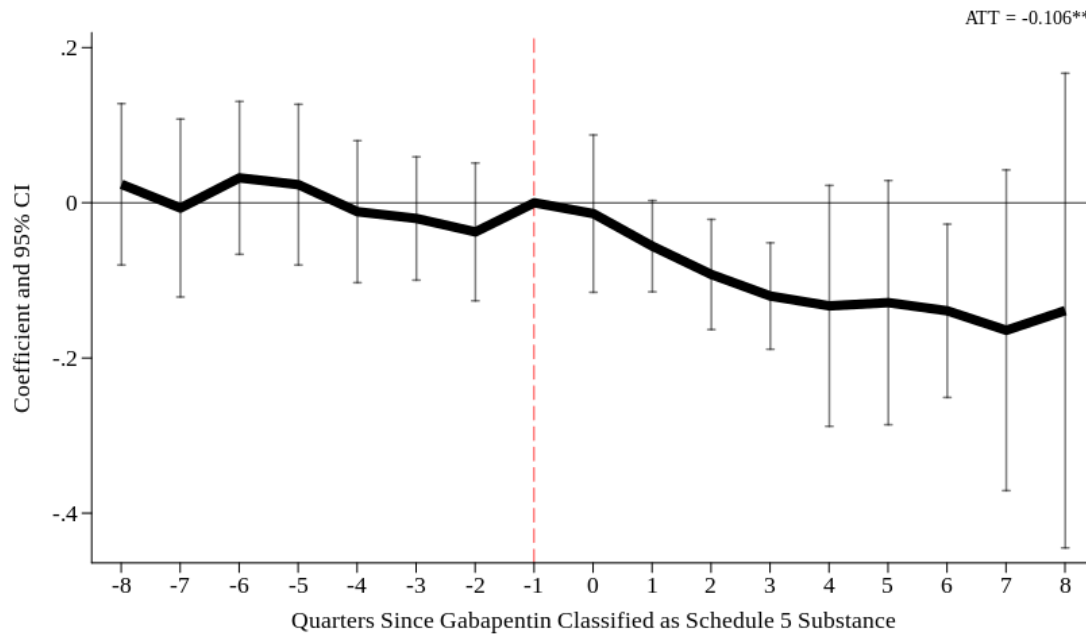


Change in rate of patients per 1,000 with a Pregabalin prescription following classification of Gabapentin in state to schedule V controlled substance.

Figure 8: Effects of Schedule V Classification on Mortality Involving Gabapentin



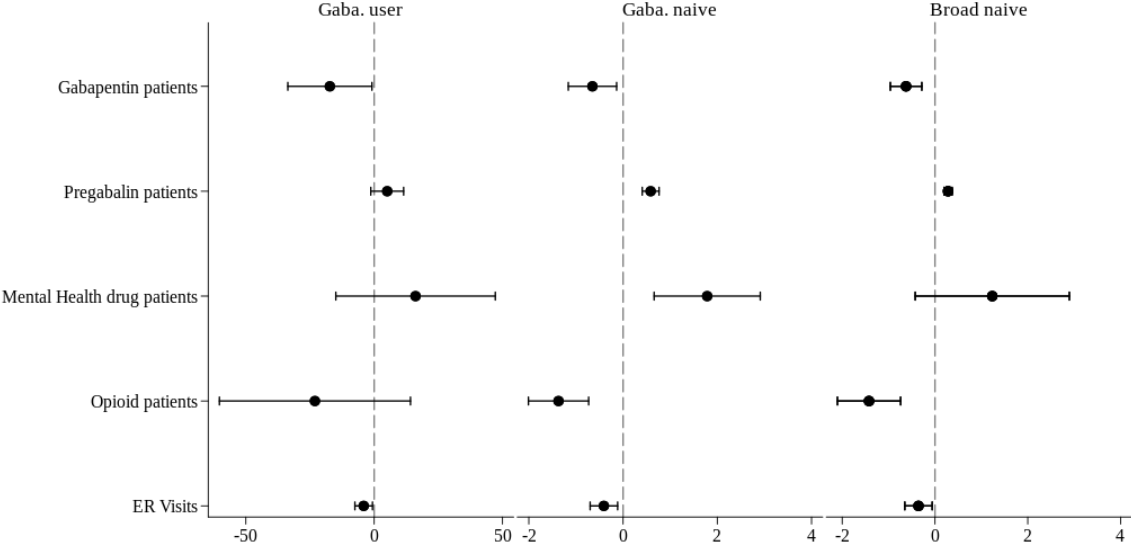
(a) Gabapentinoids



(b) Gabapentinoids + Opioids

Note: (a) Change in mortality rate per 100 000 of deaths including anticonvulsants following classification of Gabapentin in state to Schedule V controlled substance. (b) Change in mortality rate per 100 000 of deaths including anticonvulsants and opioids following classification of Gabapentin in state to Schedule V controlled substance.

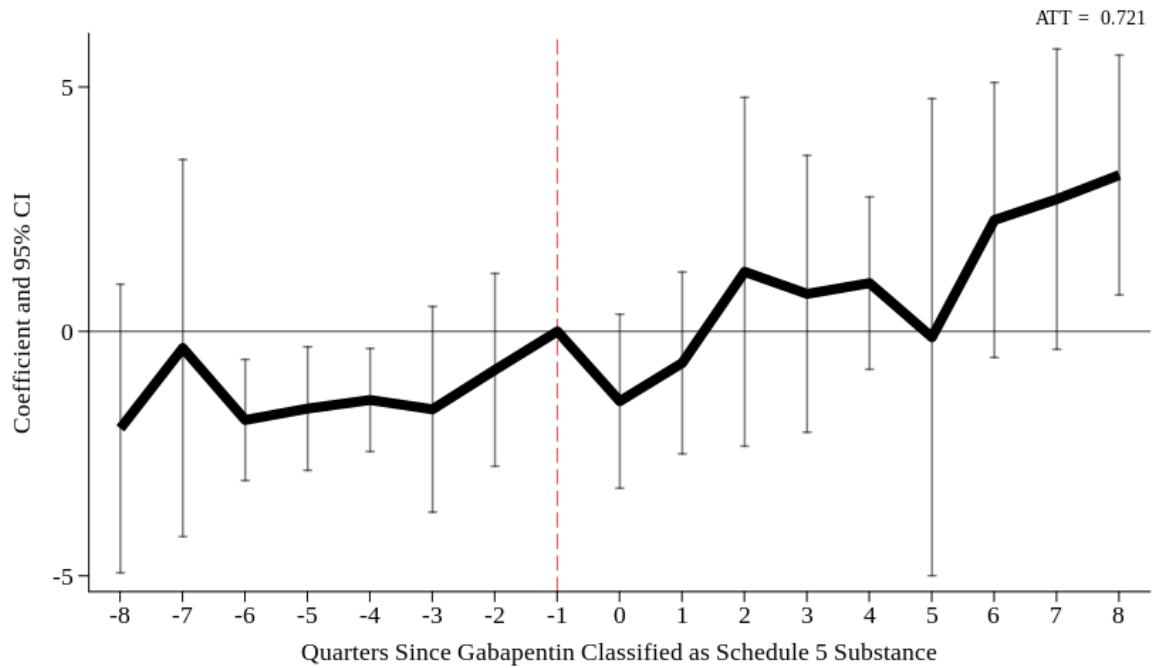
Figure 9: Schedule V Classification - Impacts by Existing Gabapentin Exposure



Note: Change in rate of patients per 1,000 gabapentin, pregabalin, mental health drug, or opioid prescription or ER visits following classification of Gabapentin in state to schedule V controlled substance, estimated for continuously enrolled patients based on 2016 Gabapentin and other medications exposure.

Policy Impacts on Non-Pharmacological Pain Management

Figure 10: Schedule V Classification - Non-Pharmacological Pain Management



Change in rate of patients per 1,000 with a non-pharmacological pain management procedure following classification of Gabapentin in state to schedule V controlled substance.

Tables

Table 1: Characteristics and Regulations: Tiers of Prescribing Restrictions.

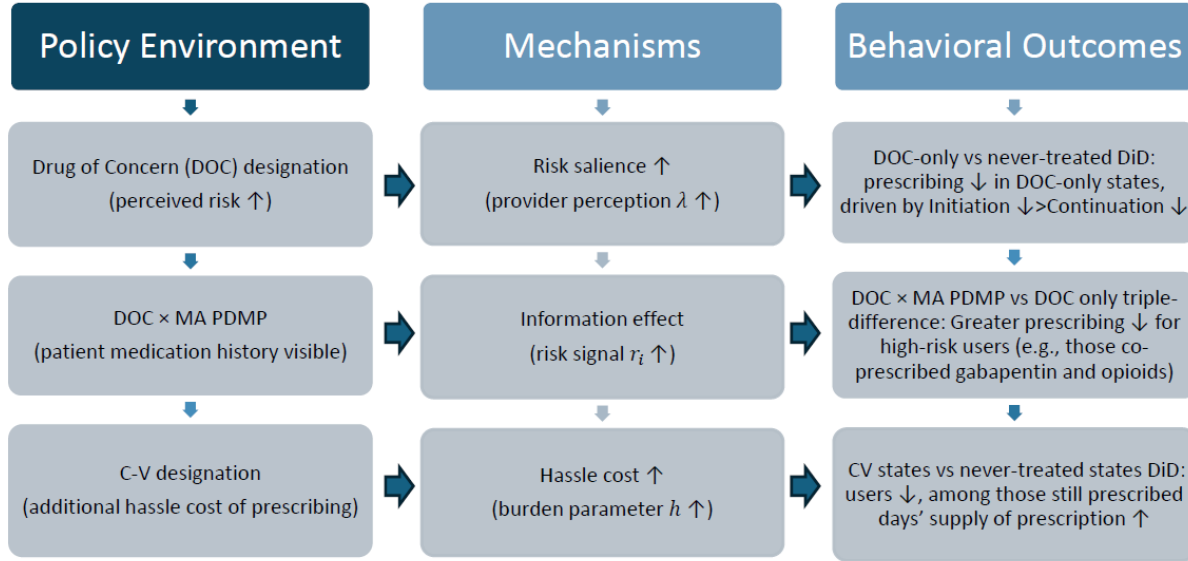
Prescription drugs	Drug of Concern	Schedule V Controlled Substance (Criterion: medical use, dependence and misuse potential)
1. Little/ no known risk of abuse. E.g. Antibiotics	1. Minimal abuse potential E.g. mild narcotic cough medicines	1. Low abuse potential E.g. Benzos, tramadol
2. Requires valid prescription	2. Prescription mostly valid for 12 mo. [†]	2. Prescription mostly valid for 12 mo. [†]
3. Electronic/ phone scripts allowed w/o patient-provider in-person contact		
4. Refill as authorized by practitioner		
5. Delegate prescribing allowed		
6. Not require prescriber DEA number	6. Other general restrictions: • Require prescriber DEA number • Prescriptions must include patient name, patient DOB, drug name, dosage, strength, days of supply and refills (if any) • Prescription bottle must carry FDA warning label prohibiting transfer • Volume restrictions on wholesale distribution • Locked storage in pharmacy • Criminal penalties for unlawful or unlicensed distribution, possession, and use (without valid prescription)	
Increasing restrictions on prescribing		

[†]With state ability to specify alternative validity limits. [‡]For instance, Kentucky mandates that prescription be filled within 60 days.

Table 2: State-Quarter Summary Statistics

	N	Mean	St. Dev.
Patients enrolled	1552	414,262.4	476579.5
Gabapentin prescriptions per 1,000 patients	1552	12.47189	3.024941
Pregabalyn prescriptions per 1,000 patients	1552	2.196201	.8060852
Opioid prescriptions per 1,000 patients	1552	24.28274	7.187883
Co-prescriptions of gabapentin and opioids per 1,000 patients	1552	3.025954	.953004
Anticonvulsant involved mortality per 100,000	2824	.3439374	.5275725
Anticonvulsant and opioid involved mortality per 100,000	2824	.2765101	.4899691
Gabapentin non-fatal poisoning diagnoses per 10,000 patients	1552	.388167	.2268058
Non-pharmacological pain management per 1,000 patients	1552	56.87691	18.99491

Table 3: Empirical Mechanisms and Testable Predictions from the Model of Prescribing Under Policy-Induced Risk Saliency and Hassle Costs



Notes: ADD

Table 4: Mechanism Check

	(1)	(2)	(3)	(4)
VARIABLES	No MA-PDMP	No MA-PDMP Gaba-Naive	No MA-PDMP Gaba-Existing	Information Effect
Drug of Concern DD	-15.35 (10.67)	0.620 (0.652)	9.962 (8.044)	0.473 (0.489)
DoC DD * MA-PDMP				-0.451 (0.734)
DoC DD * MA-PDMP * Existing				-28.14** (12.68)
Observations	2,176	928	928	2,496
R-squared	0.911	0.979	0.979	0.910

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 5: Hassle Cost Check

VARIABLES	Gaba. pts per 1,000		
	(1) Full Sample	(2) Gaba-Existing	(3) Gaba-Naive
C-V DD	-2.194** (0.890)	-17.20*** (5.732)	-0.138 (0.734)
Observations	1,920	960	960
R-squared	0.008	0.982	0.978

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Appendix

Figure A1: Gabapentinoid Involved Deaths Over Time

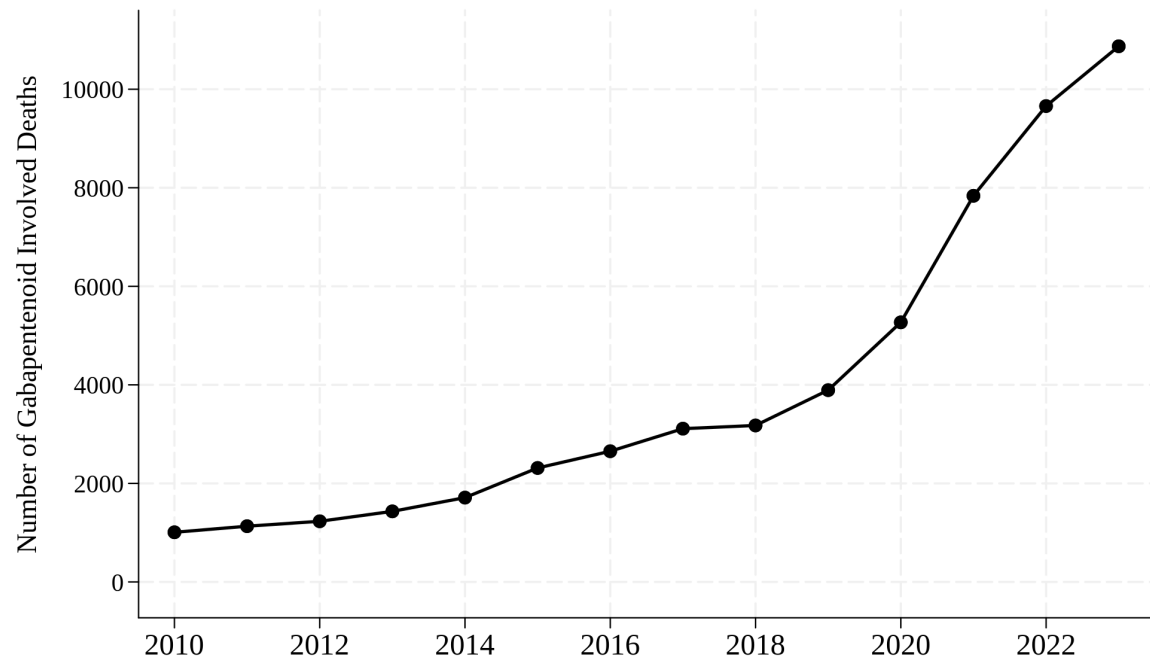
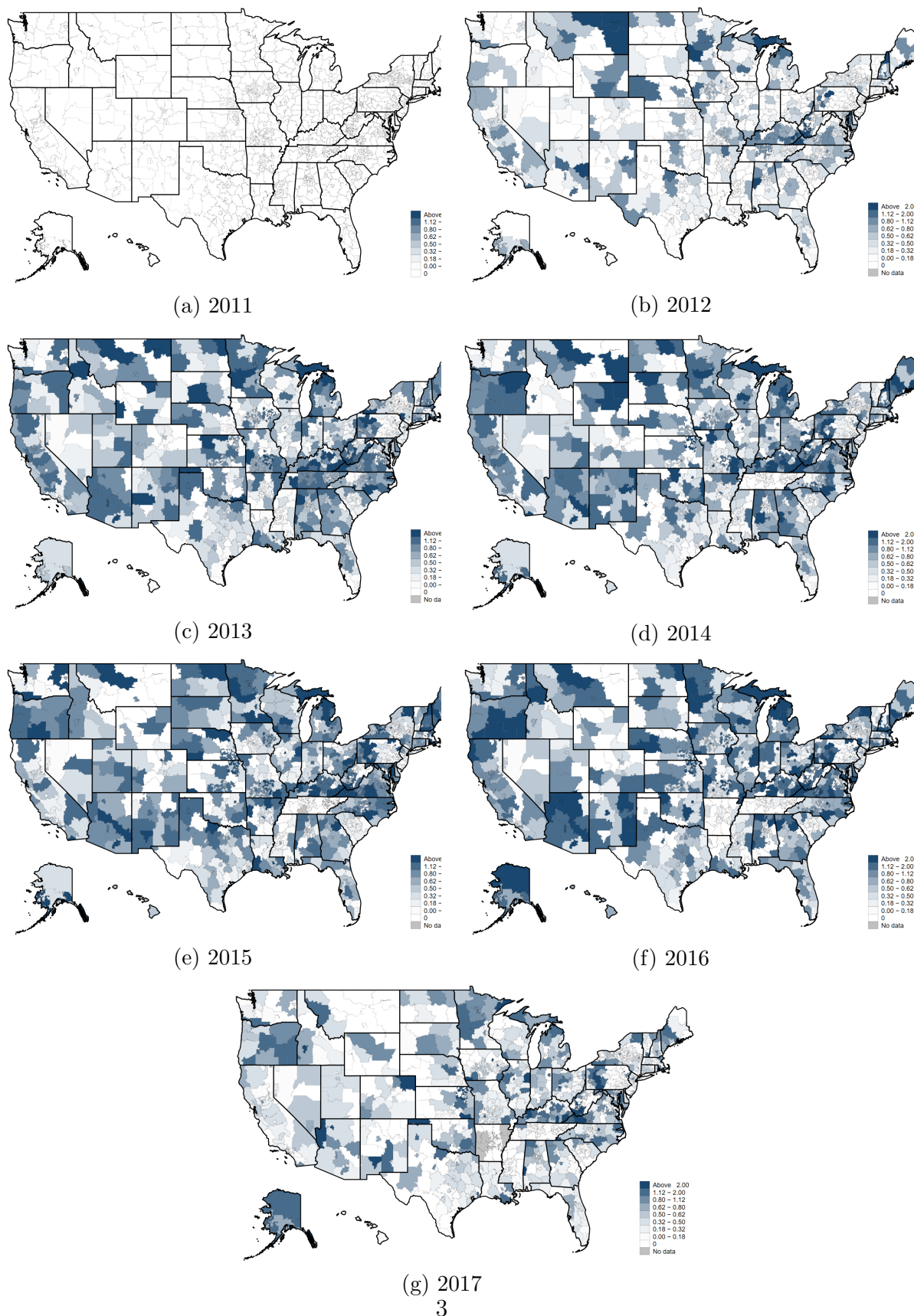
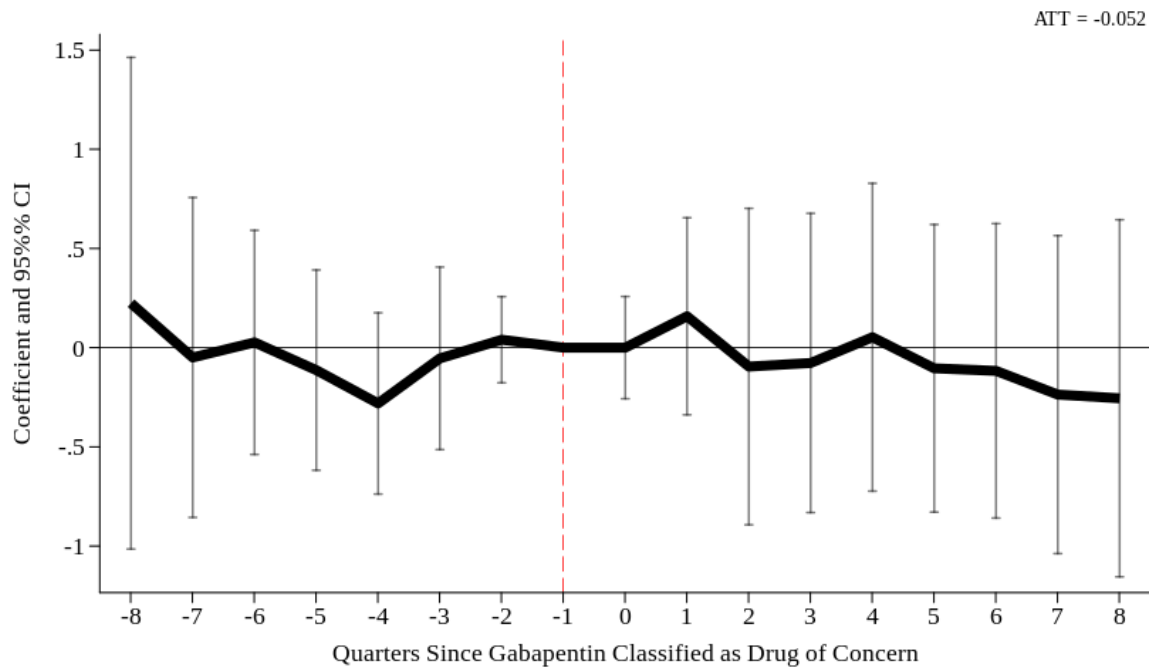


Figure A2: Poison Control Center Call Rates for Gabapentin-Involved Exposures per 100,000 Residents



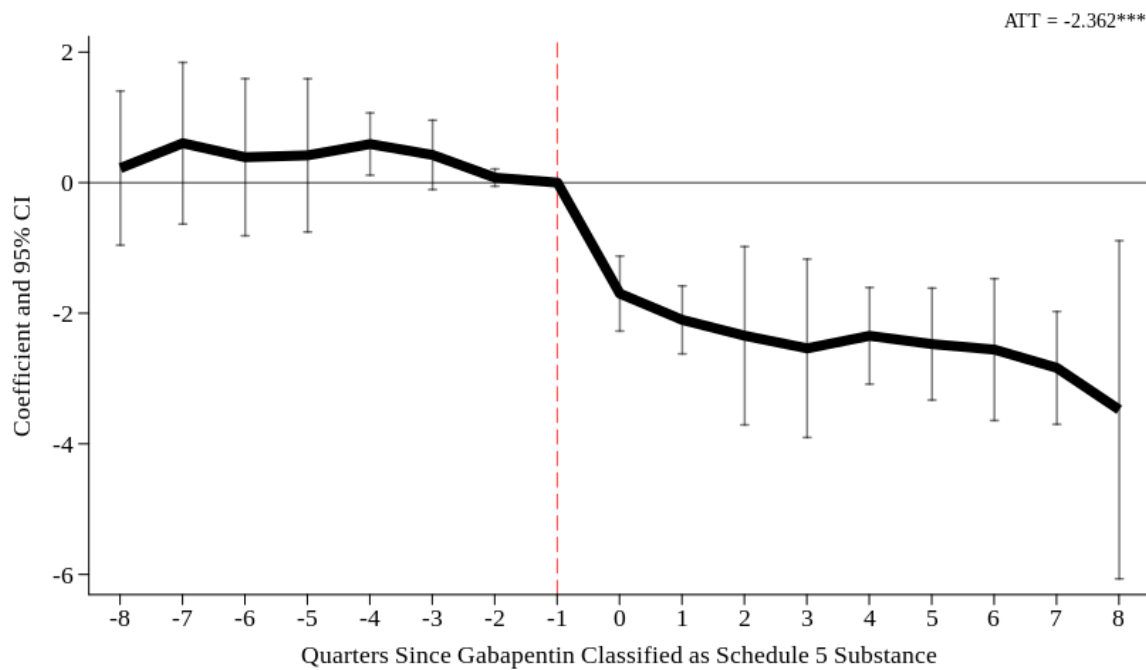
Authors' calculations based on 3-digit ZIP code-level data from the National Poison Data System (2011–2017). Percentile cutoffs are anchored to 2013 call rates. 14.63% of ZIP codes recorded no gabapentin-related poison control calls in 2013. Among ZIP codes with at least one call, the legend reflects the 10th, 25th, 40th, 50th, 60th, 75th, and 90th percentiles of call volume.

Figure A3: Drugs of Concern Classification - Gabapentin Impact (Drop Schedule V)



Notes: Change in rate of patients per 1,000 with a Gabapentin prescription following classification of Gabapentin in state to drug of concern, and thus tracked in the PDMP. This regression drops states who ever have a schedule V classification of Gabapentin.

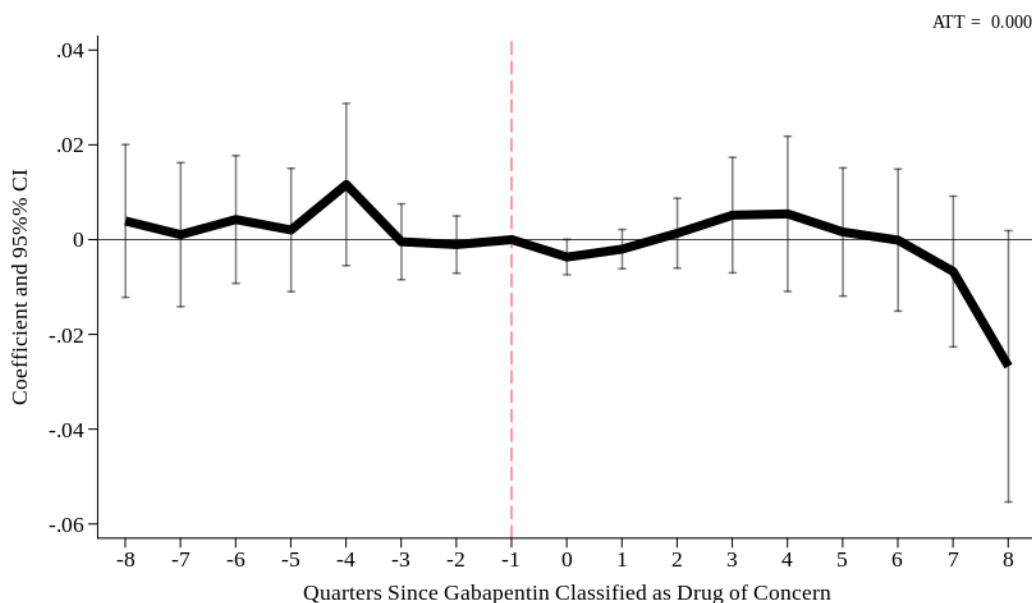
Figure A4: Schedule V Classification - Gabapentin Impact (Drop Drugs of Concern)



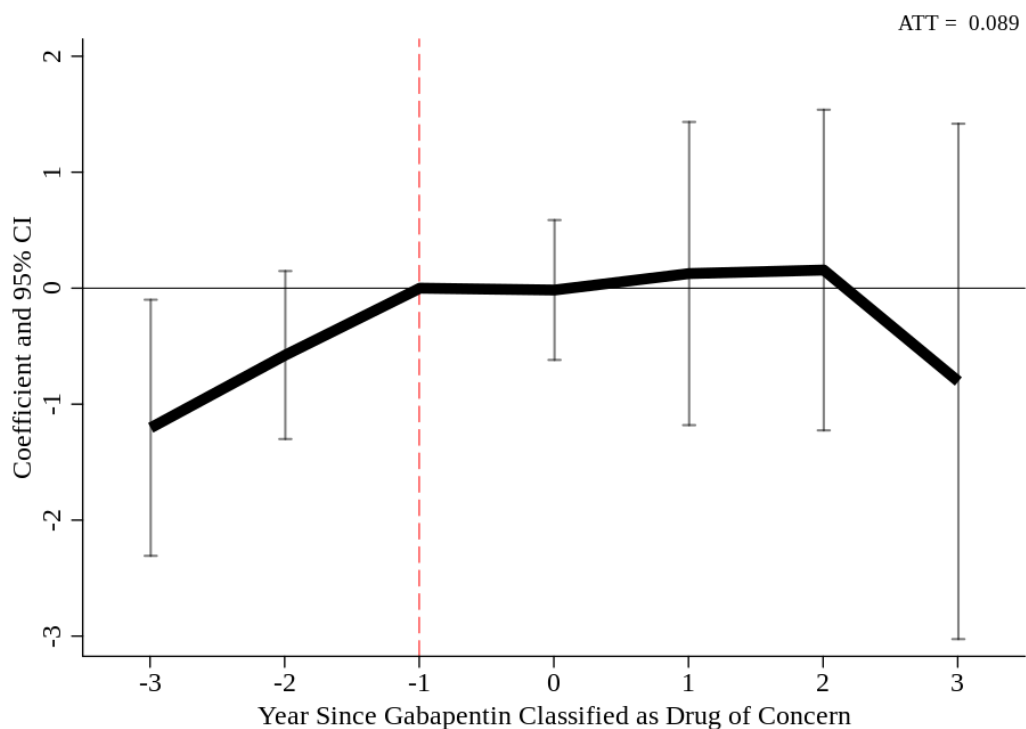
Notes: Change in rate of patients per 1,000 with a Gabapentin prescription following classification of Gabapentin in state to schedule V, and thus tracked in the PDMP. This regression drops states who only ever have a drug of concern classification of Gabapentin.

Figure A5: Gabapentin: Drug of Concern Classification - Public Payer

(a) Medicaid

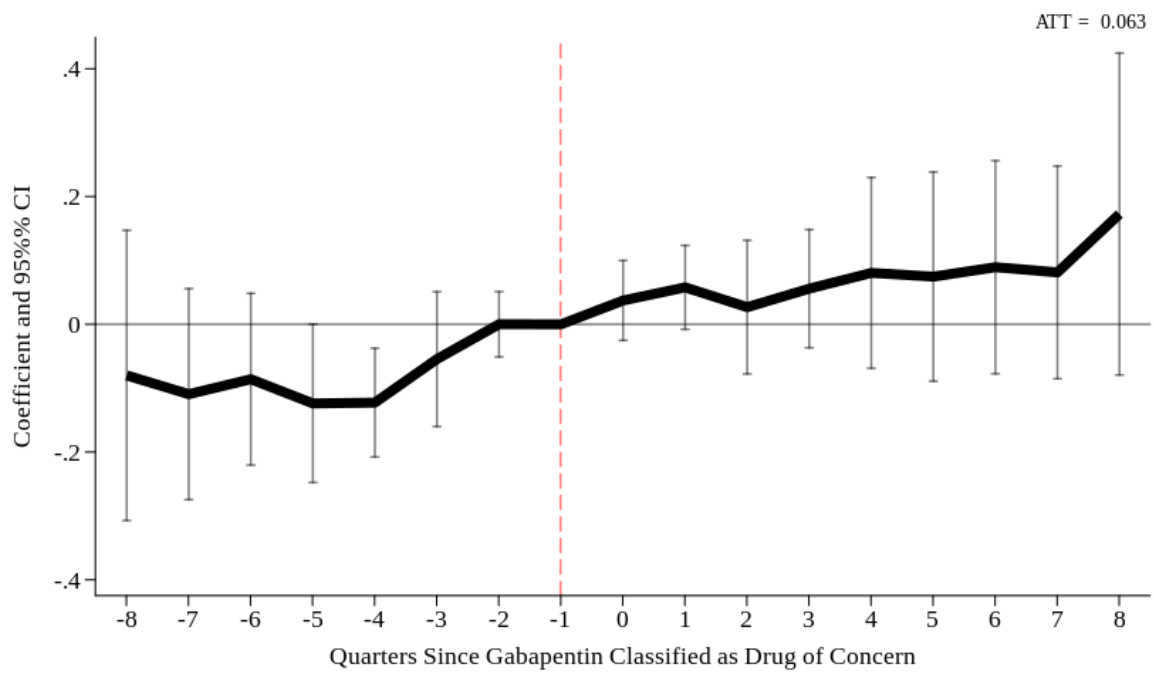


(b) Medicare



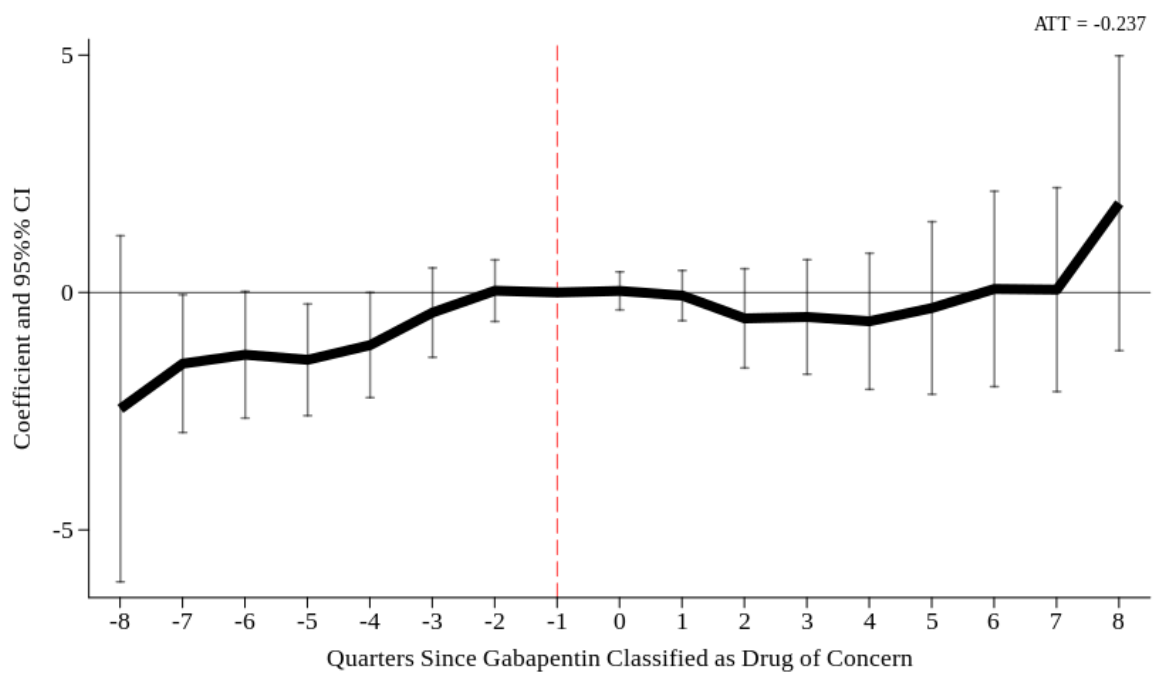
Note: (a) Change in rate of rate of Medicaid enrollees per 1,000 with a Gabapentin prescription following classification in state to drug of concern. (b) Change in rate of rate of Medicare Part D enrollees per 1,000 for an average quarter each year with a Gabapentin prescription following classification in state to drug of concern.

Figure A6: Drugs of Concern Classification - Pregabalin Impact



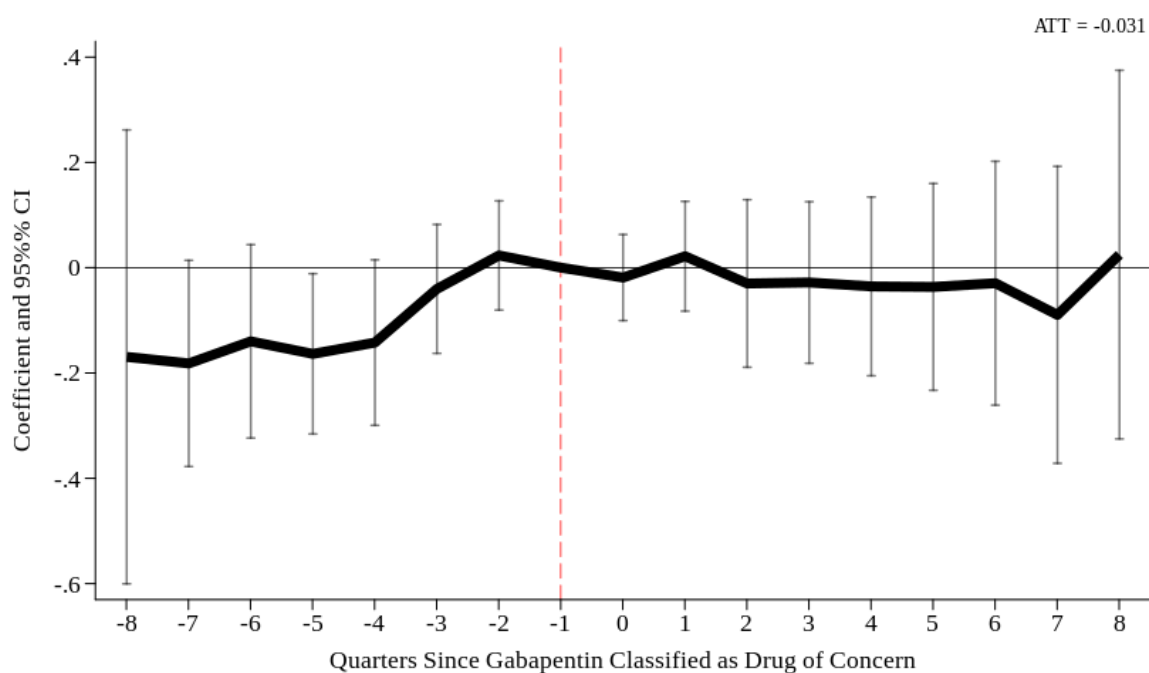
Notes: Change in rate of patients per 1,000 with a Pregabalin prescription following classification of Gabapentin in state to drug of concern, and thus tracked in the PDMP.

Figure A7: Drugs of Concern Classification - Opioid Impact



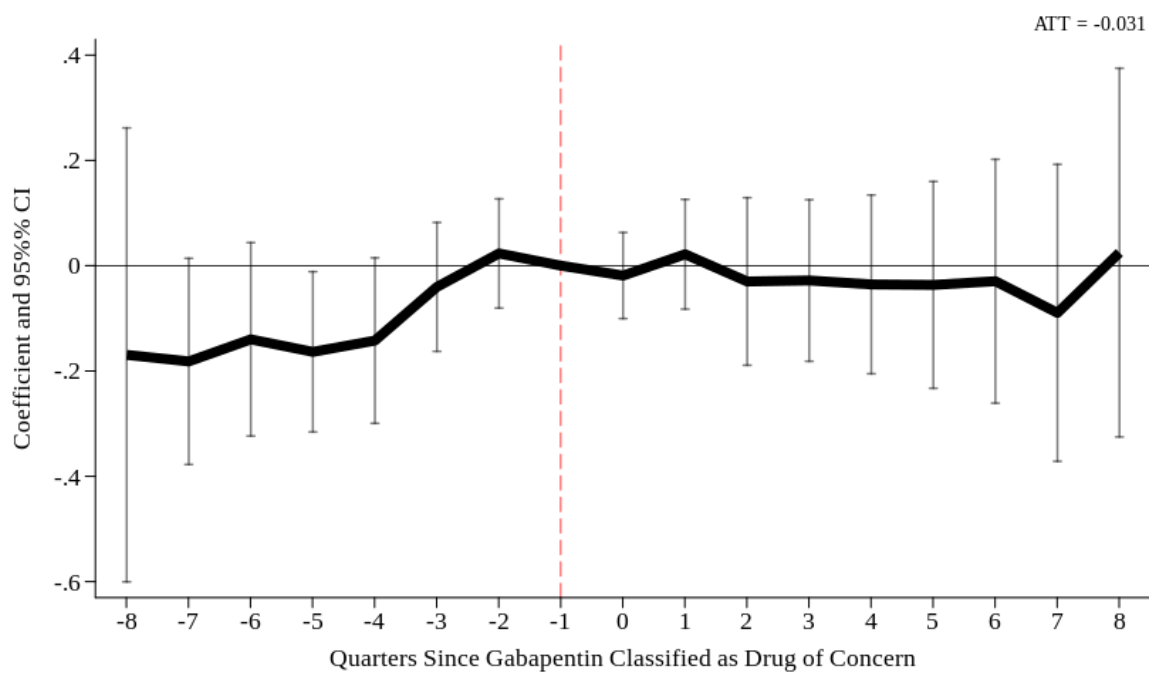
Notes: Change in rate of patients per 1,000 with a opioid prescription following classification of Gabapentin in state to drug of concern, and thus tracked in the PDMP.

Figure A8: Drugs of Concern Classification - Co prescriptions of Gabapentin and Opioids



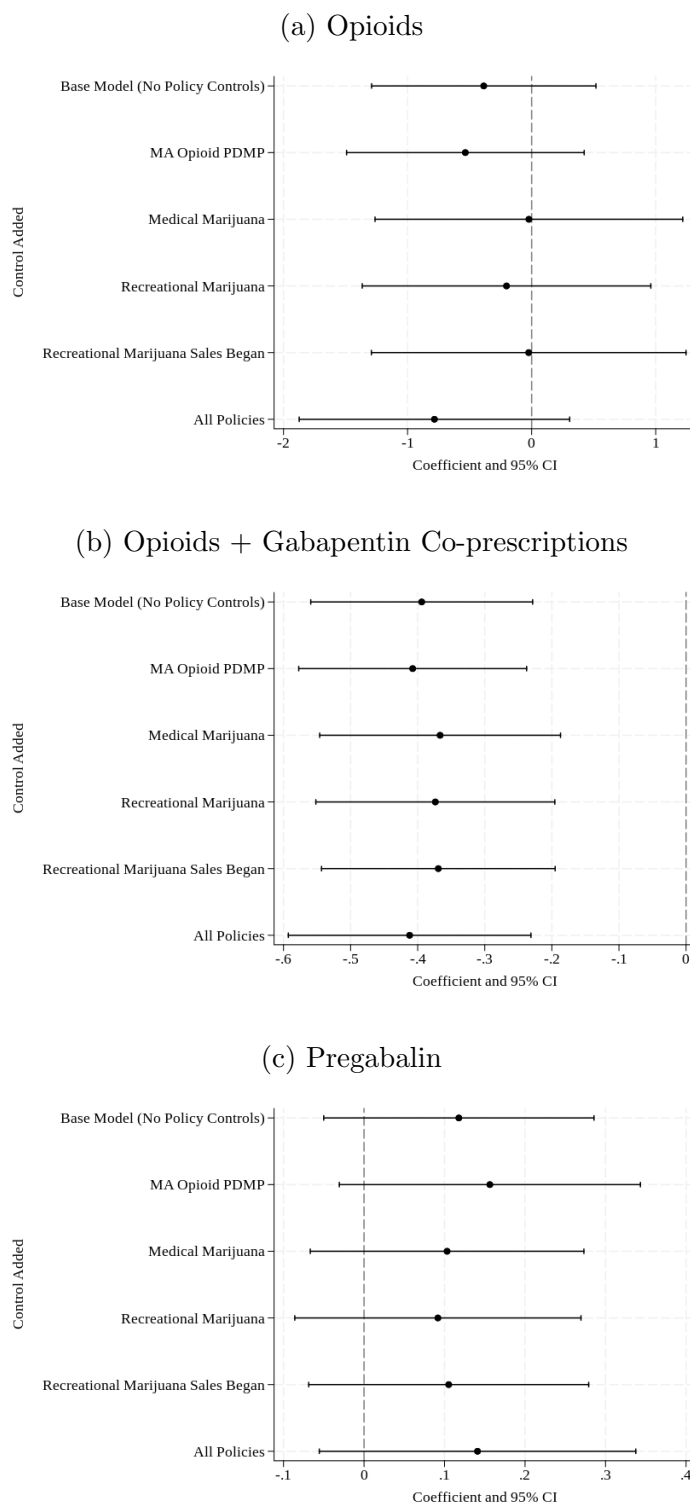
Notes: Change in rate of patients per 1,000 co-prescribed Gabapentin and an opioid following classification of Gabapentin in state to drug of concern, and thus tracked in the PDMP.

Figure A9: Drugs of Concern Classification - Co prescriptions of Gabapentin and Opioids



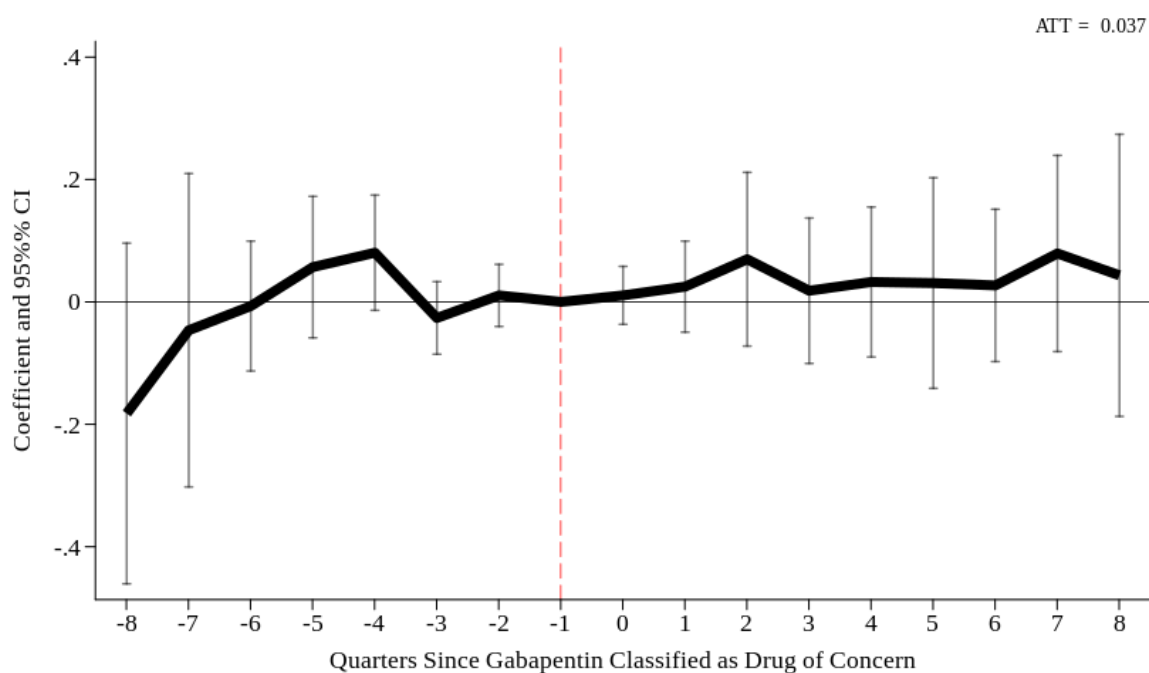
Notes: Change in rate of patients per 1,000 co-prescribed Gabapentin and an opioid following classification of Gabapentin in state to drug of concern, and thus tracked in the PDMP.

Figure A10: Potential Substitutes: Schedule V Classification Classification - Other policies Robustness Check



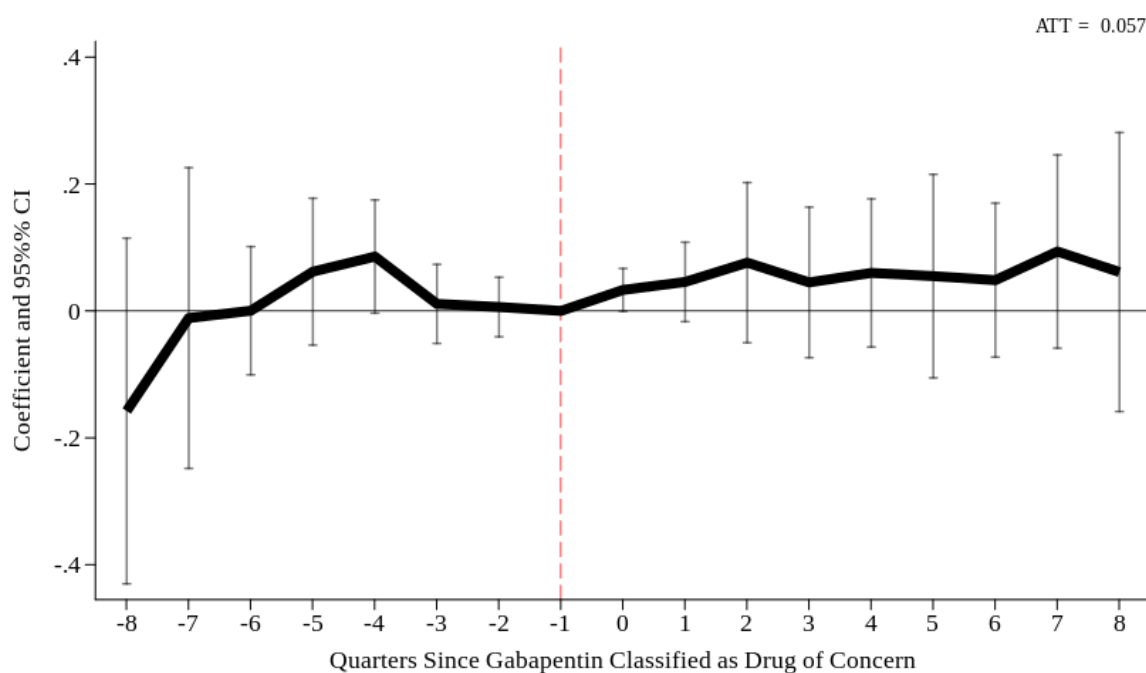
Note: Change in rate of patients per 1,000 with a (a) opioid prescription, (b) opioid and gabapentin co-prescription, and (c) pregabalin prescription following classification in state to Schedule V controlled substance. Results controlling for must access PDMP and recreational or medical marijuana policies are shown with each corresponding point estimate, with the bottom result controlling for all policies.

Figure A11: Drugs of Concern Classification - Fatal Events Involving Gabapentin



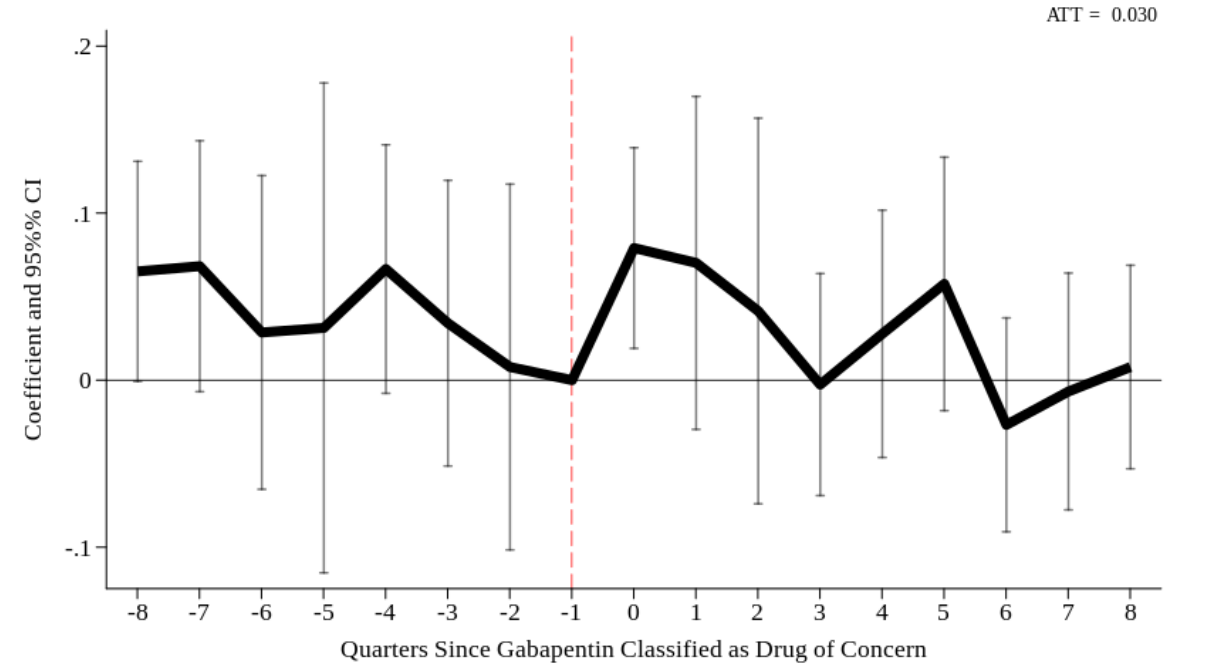
Notes: Notes: Change in mortality rate per 100,000 of deaths including anticonvulsants following classification of Gabapentin in state to drug of concern controlled substance.

Figure A12: Drugs of Concern Classification - Fatal Events Involving Gabapentin and Opioids



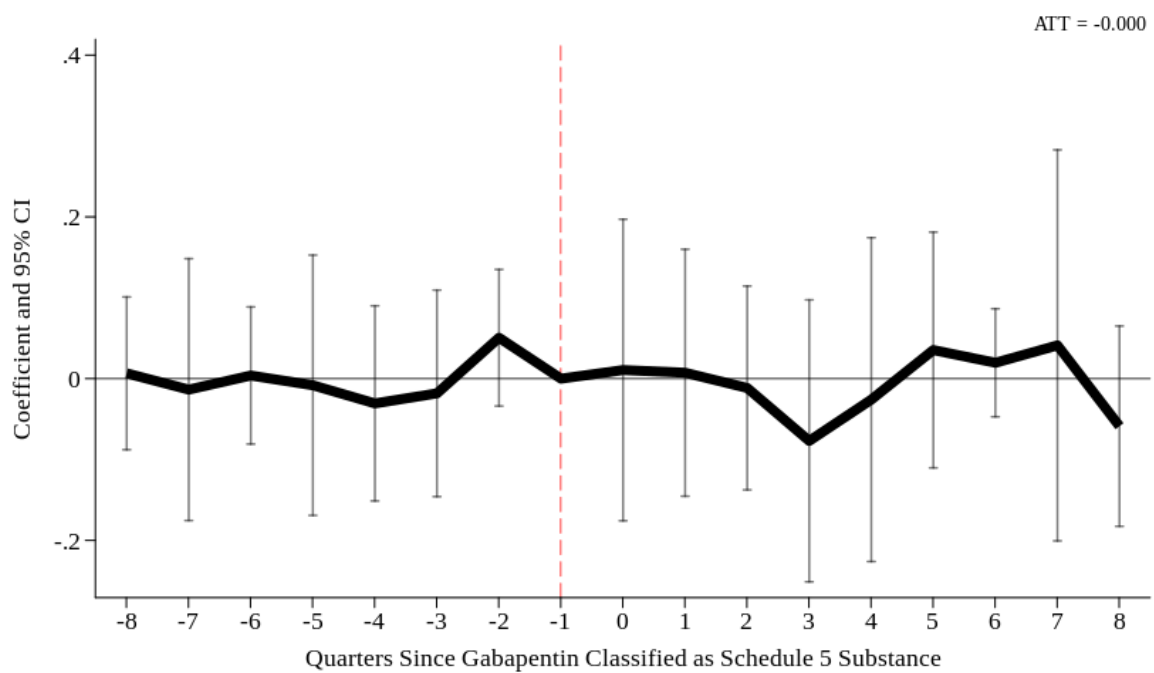
Notes: Notes: Change in mortality rate per 100,000 of deaths including anticonvulsant and opioids following classification of Gabapentin in state to drug of concern controlled substance.

Figure A13: Drugs of Concern Classification - Non-fatal Events Involving Gabapentin



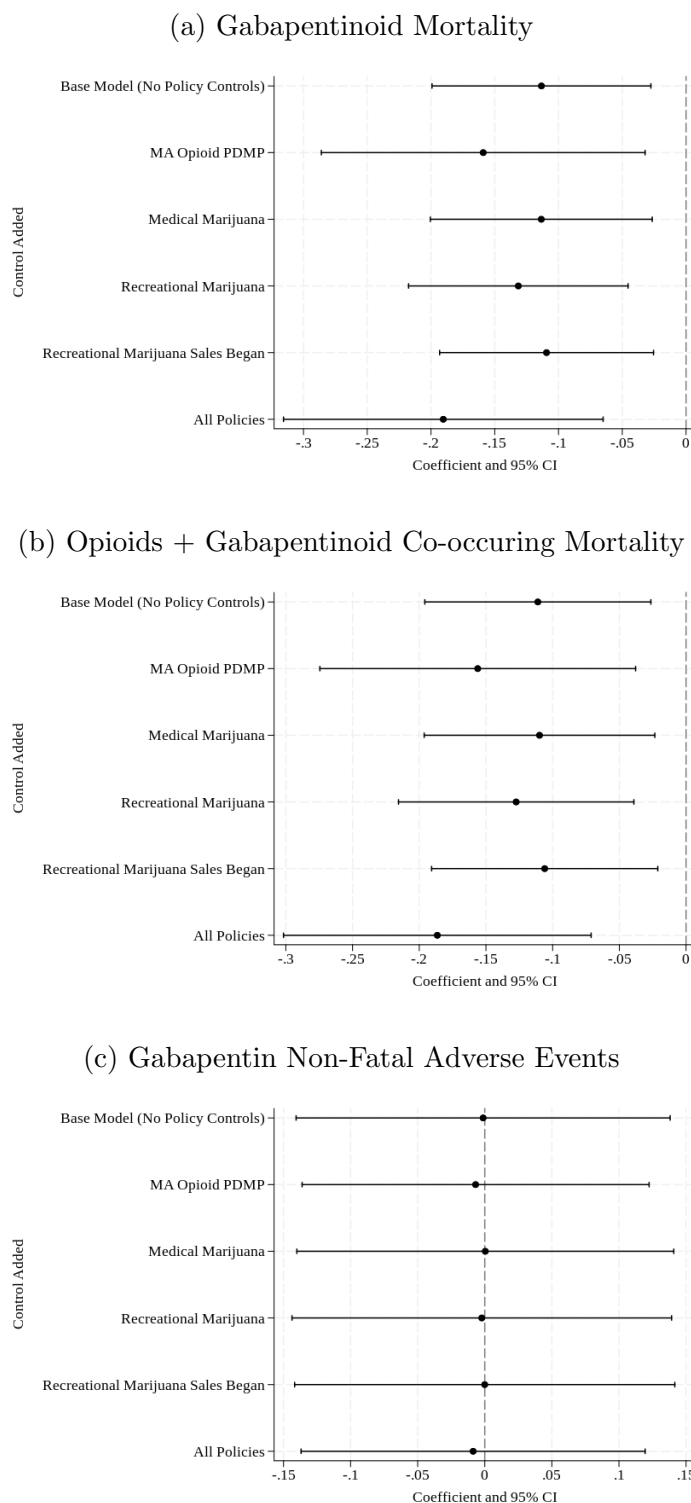
Notes: Change in incidence rate per 100,000 patients of non-fatal events including anti-convulsants following classification of Gabapentin in state to drug of concern controlled substance.

Figure A14: Schedule V Classification - Non-fatal Events Involving Gabapentin



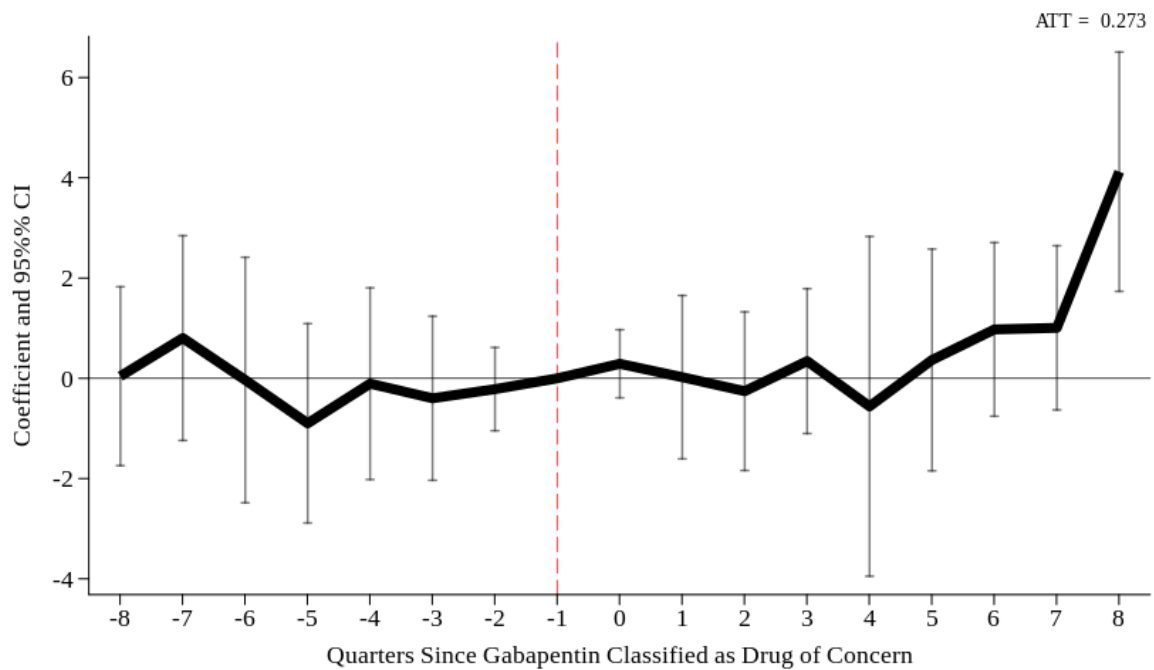
Change in incidence rate per 100,000 patients of non-fatal events including anticonvulsants following classification of Gabapentin in state to schedule V controlled substance.

Figure A15: Fatal and Non-fatal Adverse Events: Schedule V Classification Classification
- Other policies Robustness Check



Note: Change in rate of mortality per 100,000 involving (a) a gabapentinoid or (b) both a gabapentinoid and an opioid, and (c) rate per 100,00 patients of non-fatal events including anti-convulsants following classification in state to Schedule V controlled substance. Results controlling for must access PDMP and recreational or medical marijuana policies are shown with each corresponding point estimate, with the bottom result controlling for all policies.

Figure A16: Drugs of Concern Classification - Non-Pharmacological Pain Management



Notes: Change in rate of patients per 1,000 with a non-pharmacological pain management procedure following classification of Gabapentin in state to drug of concern controlled substance.

Table A1: Gabapentin PDMP reporting policies

State/District	Date	Source
Scheduled as C-V and reported to PDMP		
Alabama	11/18/19	www.alabamapublichealth.gov/blog/assets/controlledsubstanceslist.pdf
Kentucky	3/3/17	https://pharmacy.ky.gov/Documents/Gabapentin%20-%20Schedule%20V%20Controlled%20Substance.pdf
Michigan	1/9/19	https://www.michigan.gov/opioids/nel/panel-news/news/gabapentin-scheduled-as-controlled-substance-to-help-with-states-opioid-epidemic
North Dakota	4/10/19	https://nabp.pharmacy/wp-content/uploads/2016/06/North-Dakota-Newsletter-June-2019.pdf
Tennessee	7/1/18	www.tn.gov/content/dam/tn/health/healthprofboards/New%20Statue%20Gabapentin%2006-18.pdf
Utah*	5/1/24	https://dopl.utah.gov/wp-content/uploads/2024/04/Gabapentin-Fact-Sheet-1.pdf
Virginia	7/1/19	https://legacylis.virginia.gov/cgi-bin/legp604.exe?191+sum+HB2557
West Virginia	9/5/18	www.wvlegislature.gov/bill_status/bills_history.cfm?input=4336&year=2018&sessiontype=RS&btype=bill
Drug of Concern reported to PDMP		
Connecticut	1/1/21	https://portal.ct.gov/-/media/DCP/drug-control/PMP/pdf/Gabapentin.Naloxone.CPMRS.2020.pdf
District of Columbia	6/7/19	https://dchealth.dc.gov/service/prescription-drug-monitoring-program
Illinois	7/7/23	www.ilga.gov/Commission/jcar/admincode/JCARTitlePart.asp?Title=077&Part=2081
Indiana	7/1/19	www.in.gov/pla/inspect/
Kansas	5/11/18	www.ktracs.ks.gov/home/showpublisheddocument/8408/638585193066700000
Louisiana	1/1/21	https://wwwcfprd.doa.louisiana.gov/boardsandcommissions/MeetingMinutes/336_2021-1013_Minutes.FinalPkg.pdf
Massachusetts	8/1/17	www.mass.gov/doc/requirements-for-gabapentin-data-submission-0/download
Minnesota	5/31/16	www.revisor.mn.gov/laws/2016/0/185/#laws.0.1.0
Mississippi	5/7/21	https://www.mbp.ms.gov/sites/default/files/inline-images/The%20Script%20VOL%209%20July%202021-2%20%282%29.pdf
New Jersey	5/6/19	www.njconsumeraffairs.gov/Adoptions/dirado.05062019.pdf
New Mexico	2/28/23	www.nmpmp.info/wp-content/uploads/2023/11/NM-PMP-Data-Submission-Dispenser-Guide_v4.0.pdf
North Dakota	8/1/17	https://nabp.pharmacy/wp-content/uploads/2016/06/North-Dakota-Newsletter-September-2017.pdf
Ohio	12/1/16	https://www.pharmacy.ohio.gov/documents/pubs/newsletter/2019/update%20on%20gabapentin%20in%20ohio.pdf
Oregon	1/1/20	https://www.oregon.gov/oha/PH/PreventionWellness/SafeLiving/PDMP/Documents/Oregon_PDMP_Data_Submission_Dispenser_Guide.pdf
Utah	12/9/21	https://adminrules.utah.gov/public/rule/R156-37f/Current%20Rules
Virginia	2/23/17	https://legacylis.virginia.gov/cgi-bin/legp604.exe?171+ful+CHAP0181
Wisconsin	9/1/21	https://pdmp.wi.gov/statistics/controlled-substance-dispensing
Wyoming	7/1/17	https://wyoleg.gov/arules/2012/rules/ARR23-005.PDF

*Utah's schedule V classification occurred after our sample period and, thus, is considered only treated through it's original drug of concern policy.

Nebraska is excluded from the study, as their state law required the PDMP to track all prescriptions.